SkyMamba: Integrating Transformer and State-Space Model for UAV Remote Sensing RGB-D Images Semantic Segmentation

Lin Gao, Xia Li, Yu Liu, Xi Chen, Yuxiang Liu, Jiazhen Xu, and Maojun Zhang

***Abstract*—Uncrewed aerial vehicles (UAVs) aerial remote sensing (RS) images offer higher spatial resolution compared with satellite RS images, providing more detailed scene information. The flexibility of UAV-mounted sensors allows for the simultaneous acquisition of multimodal data, including RGB images and depth maps. Depth information, serving as a supplementary cue, has been extensively utilized in semantic segmentation research. Nevertheless, the current investigations on red, green, blue-depth (RGB-D) image semantic segmentation predominantly concentrate on indoor or road scenarios. In UAV RS scenes, there exist numerous objects featuring similar visual texture manifestations. The introduction of depth geometric information can efficaciously enhance the semantic segmentation accuracy of complex ground objects. Concurrently, high-resolution RS images also present challenges to the modeling of global information in large-scale space. To address these challenges, we propose the first framework for semantic segmentation of RGB-D images in UAV RS, namely, SkyMamba. This framework effectively integrates visual texture information and geometric depth information. Simultaneously, it leverages the long-sequence modeling capability of the state-space model to efficiently capture global context information in the RS space, achieving efficient semantic segmentation of RS scenes. Furthermore, we introduce a novel benchmark dataset for semantic segmentation of UAV RS RGB-D images, named comprehensive urban-scale depth, which is divided into two versions based on coarse-grained and fine-grained semantic annotation strategies to provide diverse challenges. Experimental results demonstrate that the proposed method exhibits state-of-the-art performance on both versions of the dataset, particularly in handling imbalanced scene semantic class distributions. The code for this article is available at the following URL: https://github.com/CapyLin-G/CUS-Depth**

***Index Terms*—High-resolution remote sensing (RS) images, multimodal features, red, green, blue-depth (RGB-D) image semantic segmentation, state-space model (SSM), uncrewed aerial vehicles (UAVs) RS scenario.**

I. Introduction

In RECENT years, with the widespread adoption of uncrewed aerial vehicles (UAVs), there has been a significant surge in their use for remote sensing (RS) image acquisition. Compared with satellite RS platforms, UAVs offer more flexible and controllable flight path planning, and their lower flight altitudes provide higher spatial resolution in images, while also reducing development and operational costs [1], [2]. The high-resolution RS images collected by UAVs are critical resources in various research fields, facilitating in-depth understanding and analysis in urban planning [3], environmental change monitoring [4], agricultural management [5], and disaster response [6].

In the domain of RS, the fundamental step for conducting diverse analytical studies on high-resolution images acquired from the UAV vantage point is the semantic segmentation of the spatial scene. Semantic segmentation is defined as the process of assigning each pixel in an image to a specific category, thereby partitioning the image into distinct, nonoverlapping regions, and annotating these regions with appropriate category labels [7]. This process is particularly crucial in the RS context, where the emphasis is on capturing the contextual semantic information at a large spatial scale. Such information is essential for simplifying the complex distribution of terrestrial objects within the spatial domain. The semantic segmentation of RS images has emerged as a cornerstone for numerous computer vision research endeavors within the RS community. Key applications include, but are not limited to, road extraction [8], land-use mapping [9], and urban feature recognition [10]. This foundational role underscores the significance of semantic segmentation in advancing the understanding and analysis of terrestrial landscapes through the lens of RS technology.

In semantic segmentation tasks, UAV RS images provide high-spatial resolution and capture rich semantic information from multiple altitudes and viewing angles. However, processing UAV RS images presents two key challenges. The first challenge stems from the limitations of existing red, green, blue-depth (RGB-D) semantic segmentation methods, which are primarily designed for ground-level scenarios such as autonomous vehicle navigation [11], robotic perception [12], and simultaneous localization and mapping (SLAM) [13]. These methods face significant challenges when applied to UAV RS imagery, where variations in altitude, perspective, and scene complexity introduce additional difficulties. Due to the high-spatial resolution of UAV images, complex contextual relationships exist among different objects. Unlike natural images captured at road level, UAV RS images are obtained from an aerial perspective, where ground features can appear in any orientation [14]. However, many existing methods rely on feature extraction techniques with fixed directional assumptions. For instance, convolutional neural networks (CNNs)

Received 31 March 2025; revised 1 July 2025 and 28 July 2025; accepted 3 August 2025. Date of publication 12 August 2025; date of current version 8 September 2025. This work was supported by the National Natural Science Foundation of China (NSFC) under Grant 62171451. *(Lin Gao and Xia Li contributed equally to this work.) (Corresponding author: Xi Chen.)*

Lin Gao is with the College of Systems and Engineering, National University of Defense Technology, Changsha, Hunan 410005, China, and also with AVIC Digital Corporation Ltd., Beijing 100020, China (e-mail: gaolin0914@nudt.edu.cn).

Xia Li, Yu Liu, Xi Chen, Yuxiang Liu, Jiazhen Xu, and Maojun Zhang are with the College of Systems and Engineering, National University of Defense Technology, Changsha, Hunan 410005, China (e-mail: lixia24@nudt.edu.cn; jasonyuliu@nudt.edu.cn; xi_chen@nudt.edu.cn; liuyuxiang17@nudt.edu.cn; xjz22@nudt.edu.cn; mjzhang@nudt.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3597372

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:20:32 UTC from IEEE Xplore. Restrictions apply.

primarily depend on local receptive fields, while Transformer-based architectures may be constrained by window partitioning or predefined attention patterns, limiting their ability to fully capture omnidirectional spatial relationships. This constraint hampers the comprehensive understanding of large-scale RS scenes. Consequently, the development of methods capable of extracting omnidirectional spatial features and modeling global contextual information from large-scale RS imagery is crucial for advancing dense prediction tasks in RS [15], [16].

The second challenge arises from the integration of multimodal data. With the widespread adoption of 3-D sensors, UAV imagery has expanded beyond traditional RGB images to include modalities such as point clouds and depth maps. This advancement has drawn increasing attention to the use of RGB-D data for advanced scene understanding. Depth information provides essential geometric cues regarding object relationships within a scene, effectively compensating for the limitations of RGB visual data and improving semantic segmentation accuracy [17], [18], [19], [20], [21], [22], [23], [24], [25]. Most current research on multimodal data integration focuses on RGB-D data fusion from a ground perspective, while studies on RGB-D data fusion from a drone perspective are relatively scarce. This is partly due to the limitations in the development of sensors mounted on drones, which have primarily led to the collection of image data in the RGB modality. In addition, obtaining spatially aligned RGB and depth modality data has proven to be quite challenging. As a result, there is a lack of benchmark datasets for RGB-D semantic segmentation from the drone perspective. Therefore, most existing UAV RS semantic segmentation methods mainly focus on RGB images [14], [26], while approaches tailored for UAV RS RGB-D semantic segmentation remain largely unexplored. Many objects in UAV imagery exhibit similar visual textures in RGB data while demonstrating significant differences in-depth characteristics (e.g., grasslands and high vegetation). Relying solely on RGB visual cues often leads to misclassification. Developing robust methods to effectively integrate RGB and depth information is essential for enhancing UAV RS semantic segmentation accuracy.

In light of the preceding context, we present SkyMamba, a novel framework tailored for the semantic segmentation of RS RGB-D images captured from UAVs. The cornerstone of this framework is the ground object feature extraction module (GOFEM), which incorporates an attention mechanism to synergize visual texture details and depth information within RS scenes, thereby enhancing the precision of semantic segmentation. Furthermore, to address the challenge of capturing global contextual information in expansive RS environments, the GOFEM is inspired by the strategies of Vision Mamba (Vim) [27] and VMamba [28]. It employs techniques of spatial segmentation and feature sequence reconstruction to enhance the representation of contextual information. Specifically, we subdivide the extensive RS domain into several subregions and reorder the features in designated directions. Utilizing the linear-time complexity and contextual information processing capabilities of the state-space model (SSM), we extract features from the vast RS environment in various directions, thereby enabling the representation of global spatial contextual information. Due to the design of the Mamba architecture, this module's computational complexity is reduced from $O(N^2)$ in conventional Transformers to $O(N)$ (where N denotes the sequence length). This significantly enhances computational efficiency. In addition, in RS images, adjacent objects belonging to distinct semantic categories can manifest anomalous category feature responses at their boundaries due to their similar appearances. To effectively harness the complementary information offered by multimodal data, we have incorporated a lightweight feature guidance enhancement (FGE) module at the initial stage of our framework. The multimodal feature guidance enhancement module (MFGEM) is designed to improve the quality of multimodal feature representations by effectively attenuating anomalous feature responses in peripheral regions and enhancing feature responses within the same semantic category, thereby boosting the efficiency and performance of the model during the feature encoding learning phase. By implementing these strategies, the SkyMamba framework offers a more accurate and robust solution for the semantic segmentation of RS images from a UAV perspective.

Currently, there is a lack of publicly accessible benchmark datasets tailored for the semantic segmentation of UAV RS RGB-D images. To fill this void, we have developed a novel UAV RS RGB-D image semantic segmentation benchmark dataset, named comprehensive urban-scale depth, utilizing oblique photogrammetry technology. This dataset consists of 4336 UAV RS images with a resolution of 6144×4096 pixels, accompanied by corresponding true depth data, spanning an area of 2.85 km^2 encompassing both urban and rural scenes. CUS-Depth offers annotations at two distinct granularity levels: CUS-Depth V1 includes ten semantic categories, whereas CUS-Depth V2 encompasses 18 semantic categories. We conducted experimental evaluations of SkyMamba on this dataset and compared it with six RGB-D semantic segmentation baseline methods and two RS segmentation baseline methods. The experimental results show that SkyMamba achieved a mean intersection over Union (mIoU) of 78.46% on the CUS-Depth V1 dataset and 63.82% on the CUS-Depth V2 dataset. Overall, the segmentation performance of all baseline methods has improved on these datasets.

The main contributions of this article are summarized as follows.

1) We propose a novel RGB-D semantic segmentation framework, named SkyMamba. It is the first deep learning framework specifically designed for semantic segmentation tasks of RS RGB-D images from a UAV perspective.
2) The GOFEM we designed adeptly integrates multimodal RGB-D data and incorporates an SSM to capture global context information within large-scale RS environments. The GOFEM is specifically designed for multimodal semantic segmentation in extensive RS spaces.
3) We have developed a lightweight MFGEM module at the forefront of the SkyMamba framework. The MFGEM module capitalizes on the complementary information from RGB and depth modalities to refine the feature representation of various semantic categories within the scene. Consequently, this method enhances the learning

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:20:32 UTC from IEEE Xplore. Restrictions apply.

of feature encoding without incurring additional model parameters or computational overhead.

4) We present a novel benchmark dataset for UAV RS RGB-D semantic segmentation, containing two versions with coarse and fine semantic label annotations: CUS-Depth V1 and CUS-Depth V2. SkyMamba attains state-of-the-art performance when compared with six recent RGB-D semantic segmentation baseline methods and two RS segmentation baseline methods on both versions of the dataset, demonstrating its effectiveness in addressing the challenge of efficient segmentation in scenarios with imbalanced semantic class distributions.

The remainder of this article is organized as follows. Section II provides a brief overview of the related research status concerning the direction and methodology of this study. Section III elaborates on the proposed network model architecture and the design details of its components. Subsequently, in Section IV, we validate the effectiveness and superiority of the proposed method through extensive experiments and analysis. Finally, in Section V, we summarize the work presented in this article.

II. Related Work

A. RGB-D Semantic Segmentation

With the evolution of depth sensors and the exploration of multimodal data, depth modality data is increasingly being employed to augment RGB data in semantic segmentation tasks, a practice referred to as RGB-D semantic segmentation [29], [30], [31]. In the domain of RGB-D semantic segmentation, a prevalent approach involves utilizing depth data as an auxiliary input to refine the feature representations derived from RGB images. Long et al. [32] explored this concept by straightforwardly combining the final score maps from both RGB and depth sources to enhance the discrimination of various objects in indoor environments. Wang et al. [33] and Hu et al. [34] have innovatively extended the feature calibration mechanism, moving it beyond the final output stage of the dual-path network to multiple critical layers within the network's backbone. This improvement strategy encourages the network to perform cross-modal feature fusion at multiple stages, thereby achieving more precise feature recalibration.

To enhance the integration of feature information across RGB and depth modalities, most approaches utilize dual parallel backbone networks to independently extract features, followed by dedicated modules for multimodal feature fusion. For example, Chen et al. [35] presented a bidirectional cross-modal feature propagation technique with split and aggregation gates for RGB-D semantic segmentation. This approach effectively calibrates RGB feature responses and depth feature responses, achieving enhanced segmentation results by alternately aggregating these calibrated representations at multiple stages. In recent advancements, Yin et al. [36] introduced a novel RGB-D pretraining framework named DFormer, designed to learn transferable representations for RGB-D segmentation tasks. This framework is specifically designed to encode RGB and depth information, addressing the prevalent yet unresolved issue in existing methods where depth maps' 3D geometric relationships are mismatched during encoding.

B. UAV RS Images Semantic Segmentation

Semantic segmentation of UAV RS images offers viable solutions for visual perception in autonomous UAV navigation within unfamiliar environments [37], [38]. To enhance semantic segmentation performance, deep learning-based architectures for UAV RS image semantic segmentation have become prevalent. Deep learning-based methods can be classified into CNN-based methods, Vision Transformer-based methods, Graph-based methods, and GAN-based methods.

CNN-based methods effectively capture contextual semantic details while minimizing spatial dimensions, facilitating feature extraction and prediction generation. Xia et al. [39] employed the fully convolutional network (FCN) [32] to delineate fragmented region boundaries in UAV RS images, subsequently integrating these fragments using the oriented watershed Transform and ultrametric contour map (OWT-UCM) algorithm. Nonetheless, the smoothing induced by max-pooling layers led to discontinuous boundary connections. To mitigate this, their follow-up study [40] eliminated max-pooling layers, thereby enhancing boundary clarity in predictions. SegNet [41] incorporates a skip-connection framework to facilitate a symmetrical encoder–decoder configuration, augmenting global context by progressively decoding multiscale feature maps. It also utilizes max-pooling indices from encoder stages to prevent semantic ambiguity in predictions.

Visual Transformer-based methods commonly utilize multihead attention models within an encoder–decoder framework for the semantic segmentation of UAV RS images, mirroring the foundational structure of CNN-based approaches. The Transformer encoder is instrumental in capturing global feature relationships within the image, rendering it particularly adept for large-scale scene analysis in UAV RS. Zhang et al. [42] proposed a feature fusion method grounded in the Visual Transformer for UAV RS images based on full-scale feature aggregation and grouped feature reconstruction to capture large-scale features. In addition, other works have also proposed how to effectively aggregate and utilize information from multiscale features for UAV images and video data [43], [44], [45], [46], [47]. Yi et al. [48] developed a semantic segmentation network for urban scenes termed UAVformer, leveraging the Transformer architecture. The network incorporates a primary encoder and multiple auxiliary encoders to form a composite network backbone, enabling it to perform efficient semantic segmentation across various scales and in RS environments characterized by complex backgrounds.

Graph-based methods capture contextual semantic information in RS by establishing random dependencies between pixels or regions. Diao et al. [49] proposed a superpixel-based attention graph neural network for semantic segmentation of high-spatial resolution aerial images, constructing a K-nearest neighbor (KNN) graph for each image, where each node corresponds to a superpixel in the image and is associated with a hidden representation vector. Relying on attention mechanisms and recurrent functions, each node updates its hidden representation based on its current state and information from its neighbors, with the final representation of each node used to predict the semantic category of each superpixel. Oyang and Li [50] proposed using simple linear iterative

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:20:32 UTC from IEEE Xplore. Restrictions apply.

clustering (SLIC) to segment images into ground objects (or superpixels) and obtain features for each object from the feature maps of the upsampling layer. These features serve as the characteristics of the nodes in the graph convolutional network (GCN), effectively combining the strengths of deep learning and GCN to enhance the performance of RS image semantic segmentation.

GAN-based methods aim to reduce domain shift between the source and target domains, enabling the conversion of source domain images to the style of the target domain, thereby enhancing semantic segmentation performance on the target domain. Zan et al. [51] proposed a gradual adversarial training (GAT) approach for RS image segmentation. This method incorporates a domain adaptation mechanism, dynamically modulates input data, effectively reduces adversarial perturbations, improves segmentation accuracy, and demonstrates significant robustness against adversarial attacks. Zeng et al. [52] introduced DS-DWTGAN, an unsupervised domain adaptation method specifically designed for semantic segmentation of RS images across domains. This approach combines discrete wavelet transform (DWT) with spatial-channel modules in the generative network to simultaneously preserve semantic content from the source domain and capture rich frequency domain information, thereby reducing interdomain discrepancies.

C. State-Space Model

SSM has been influenced by applications in the control domain, employing continuous space states to capture long-range dependencies through intricate temporal dynamics. The linear state-space layer (LSSL) [53] showcased the potential of SSM in addressing long-range dependencies but also highlighted significant challenges in computational and memory efficiency. Gu et al. [54] introduced the Structured state-space sequence (S4) model, which utilizes a standard parameterization strategy to mitigate the computational burden associated with SSM. Subsequently, various variants of the S4 model have been developed to balance model performance and computational efficiency. Gu et al. [55] and Gupta et al. [56] incorporated complex diagonal structures into the S4 model to enhance the temporal modeling capabilities of SSM.

SSM, initially developed for unidirectional data sequences, is not directly applicable to image data due to the lack of a specific orientation. To overcome this limitation, researchers have incorporated a selective scanning mechanism in various directions to harness SSM's robust contextual modeling capabilities for visual research, thereby facilitating global modeling of image feature information. In recent years, the selective scanning mechanism of Mamba [57] has garnered significant attention in the field of deep learning due to its capability to model complex temporal dynamics in long sequences, thereby revolutionizing the approach to handling long-range dependencies and data processing [53], [54]. Zhu et al. [27] introduced Vim, the pioneering model that extended Mamba to the visual research domain, which selectively scans images horizontally, enabling features distributed horizontally to perceive global information. Building upon Vim, Liu et al. [28] developed VMamba, which added selective scanning in the vertical direction, thereby enhancing the model's global receptive field across different dimensions. RS images, distinct from natural images, exhibit large-scale spatial features in multiple directions. To address this, Zhu et al. [58] proposed a semantic segmentation framework for high-resolution RS images named Samba. This study represents the first demonstration of Mamba's efficacy in RS image semantic segmentation, establishing a new performance benchmark for Mamba-based technologies in this specific application.

III. Method

A. Overall Architecture

SkyMamba adheres to the conventional encoder–decoder framework for semantic segmentation of UAV RS images. Within the encoder component, multiple GOFEMs are utilized to capture and integrate visual texture features and depth features of ground objects across various scales from an aerial perspective. These multiscale features are subsequently converted into predicted ground object semantic labels via a streamlined decoder.

Given a dataset comprising UAV-captured RGB images and their corresponding depth maps, we initially utilize a MFGEM to harness the complementary information from both modalities. This module is designed to refine ground object features across different semantic classes within the scene space. It effectively mitigates aberrant feature responses in edge regions while amplifying those within the same semantic class. Following this, the refined RGB and depth features are processed through a multiscale encoder, which incorporates four GOFEMs. Each module encodes the RGB-D features at multiple scales corresponding to 1/4, 1/8, 1/16, and 1/32 of the original image dimensions. Ultimately, the encoded features at various scales are fused and input into a lightweight decoder to predict semantic labels for ground objects. The complete SkyMamba network architecture is depicted in Fig. 1, with detailed descriptions of the MFGEM (Section III-B), GOFEM (Sections III-C–III-E), and the lightweight decoder (Section III-F) provided below.

B. Multimodal FGE Module

In outdoor RS applications, objects often exhibit similar appearances yet vary significantly in height (e.g., grasslands versus tall vegetation) or share similar heights but differ markedly in appearance (e.g., roads versus grasslands). In the field of semantic segmentation, we aim for features of the same object category to be similar, while features of different categories should exhibit clear differences. The MFGEM, illustrated in Fig. 2, leverages high-confidence activations from one modality to refine and guide the features of another prior to encoding RGB and depth features. This module enhances the intracategory feature similarity and intercategory feature dissimilarity, thereby improving the overall performance of the semantic segmentation task.

1) Homogeneous Feature Aggregation (HFA): The information provided by RGB and depth modalities is complementary, as they respectively offer visual texture information and geometric structural information of objects within RS environments. To aggregate homogeneous features within objects of the same semantic category in outdoor environments, it is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:20:32 UTC from IEEE Xplore. Restrictions apply.

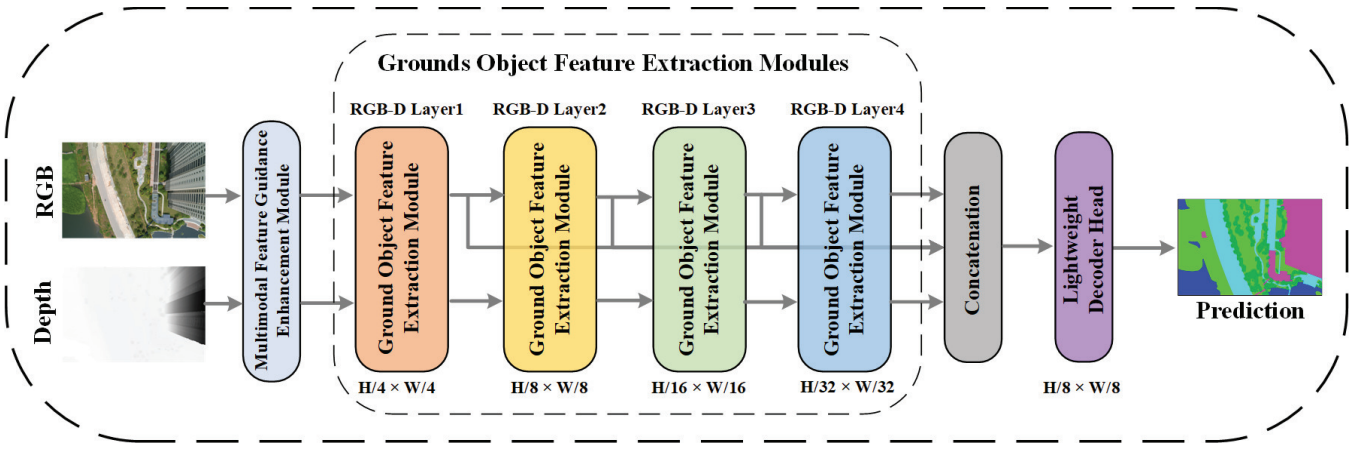


Fig. 1. Overall architecture of SkyMamba. Initially, RGB and depth modality data are processed through a lightweight MFGEM to calibrate the features of various ground objects. Subsequently, the calibrated RGB-D modality data are fed into four GOFEMs for encoding. Finally, the encoded features are transmitted to a lightweight decoder head for the final semantic prediction.

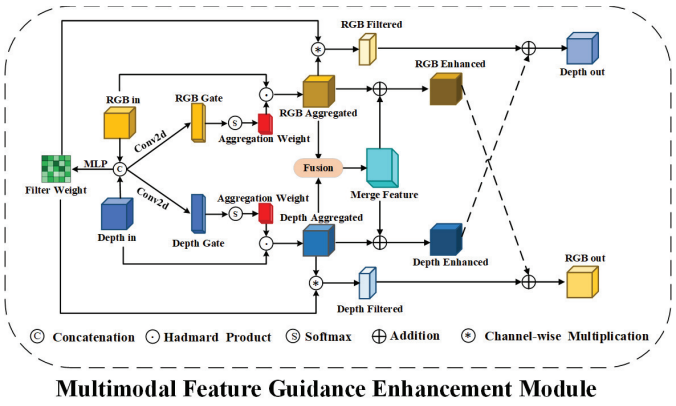


Fig. 2. Overall architecture of the MFGEM. It incorporates both homogeneous feature aggregation and heterogeneous feature separation operations. It leverages high-confidence activations from diverse modal features to guide the refinement of object features across various semantic categories. This approach effectively suppresses anomalous feature responses in the peripheral regions of category objects while reinforcing homogeneous feature responses within the same category objects.

essential to harness the representational strengths of various modalities for feature calibration. This involves integrating multimodal features to enhance the representation of objects sharing the same semantic category at corresponding spatial locations. Specifically, we concatenate the RGB and depth feature maps along the channel dimension and apply 1×1 convolution operations to generate the corresponding spatial gates G_{rgb} and $G_{\text{depth}} \in \mathbb{R}^{2C\times H\times W}$

$$F_{\text{cat}} = \text{Concat}\left(F_{\text{in}}^{\text{rgb}}, F_{\text{in}}^{\text{depth}}\right), F_{\text{cat}} \in \mathbb{R}^{2C\times H\times W} \tag{1}$$

$$G_{\text{rgb}} = \text{Softmax}\left(\text{Conv}_{1\times 1}\left(F_{\text{cat}}\right)\right) \tag{2}$$

$$G_{\text{depth}} = \text{Softmax}\left(\text{Conv}_{1\times 1}\left(F_{\text{cat}}\right)\right). \tag{3}$$

These gates regulate the information flow of the RGB and depth modal feature maps. Following this, we fuse the modal spatial gates to derive the HFA weights W_{rgb} and W_{depth}

$$W_{\text{rgb}} = \frac{e^{G_{\text{rgb}}}}{e^{G_{\text{rgb}}} + e^{G_{\text{depth}}}} \tag{4}$$

$$W_{\text{depth}} = \frac{e^{G_{\text{depth}}}}{e^{G_{\text{rgb}}} + e^{G_{\text{depth}}}} \tag{5}$$

where the HFA weights W_{rgb} and $W_{\text{depth}} \in \mathbb{R}^{1\times H\times W}$, G_{rgb} and G_{depth} represent the spatial gates for RGB and depth modality features. The HFA weights W_{rgb} and W_{depth} sum to 1 at the same position (i, j)

$$W_{\text{rgb}}^{(i,j)} + W_{\text{depth}}^{(i,j)} = 1. \tag{6}$$

The HFA weights can effectively consolidate object features of the same semantic category from both RGB and depth modalities. The input features $F_{\text{in}}^{\text{rgb}}$ and $F_{\text{in}}^{\text{depth}}$ are aggregated homogeneously to produce $F_{\text{aggregated}}^{\text{rgb}}$ and $F_{\text{aggregated}}^{\text{depth}} \in \mathbb{R}^{C\times H\times W}$

$$F_{\text{aggregated}}^{\text{rgb}} = W_{\text{rgb}} \odot F_{\text{in}}^{\text{rgb}} \tag{7}$$

$$F_{\text{aggregated}}^{\text{depth}} = W_{\text{depth}} \odot F_{\text{in}}^{\text{depth}} \tag{8}$$

where $\odot$ is used for elementwise multiplication (Hadamard product).

Subsequently, we combine the homogenously aggregated RGB and depth features through interaction to obtain the merged feature F_{merge}

$$F_{\text{merge}} = F_{\text{aggregated}}^{\text{rgb}} + F_{\text{aggregated}}^{\text{depth}}. \tag{9}$$

Finally, the merged feature F_{merge} is further utilized to enhance the two modality features, resulting in $F_{\text{enhanced}}^{\text{rgb}}$ and $F_{\text{enhanced}}^{\text{depth}} \in \mathbb{R}^{C\times H\times W}$, thereby reinforcing the similarity of features within the same category

$$F_{\text{enhanced}}^{\text{rgb}} = \frac{F_{\text{aggregated}}^{\text{rgb}} + F_{\text{merge}}}{2} \tag{10}$$

$$F_{\text{enhanced}}^{\text{depth}} = \frac{F_{\text{aggregated}}^{\text{depth}} + F_{\text{merge}}}{2}. \tag{11}$$

2) Heterogeneous Feature Separation (HFS): In outdoor scenes, there are numerous objects with visually similar textures, such as high vegetation and grass. During feature aggregation operations, these objects can easily produce ambiguous, abnormal feature responses at their edges due to their similar textures. However, high vegetation and grass exhibit significant differences in spatial geometric structures.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:20:32 UTC from IEEE Xplore. Restrictions apply.

Therefore, we consider utilizing the differences in multimodal data features to separate heterogeneous feature responses at the edges of different categories, thereby making the features distinguishing between different classes more pronounced. In practice, we employ high-confidence activation of one modality's features to filter out anomalous features at the edges of objects in another modality, thereby enhancing the contrast of edge features between different semantic categories and achieving the separation of heterogeneous features. Initially, we fuse the input RGB modality features and depth modality features, and through a multilayer perceptron (MLP) operation, we obtain the HFS weights $W_{\text{filter}} \in \mathbb{R}^C$

$$W_{\text{filter}} = \alpha\left(LN\left(GP\left(\left[F_{\text{in}}^{\text{rgb}}, F_{\text{in}}^{\text{depth}}\right]\right)\right)\right) \tag{12}$$

where $F_{\text{in}}^{\text{rgb}} \in \mathbb{R}^{C\times H\times W}$ and $F_{\text{in}}^{\text{depth}} \in \mathbb{R}^{C\times H\times W}$ represent the input RGB and depth modality features. The notation $[\cdot,\cdot]$ denotes the cross-modal feature fusion operation, *GP* stands for global average pooling, *LN* represents the MLP network operation, and α is a sigmoid function that scales the HFS weights to the range (0, 1). Subsequently, we perform channelwise multiplication between the homogenized RGB feature $F_{\text{aggregated}}^{\text{rgb}}$ and depth feature $F_{\text{aggregated}}^{\text{depth}}$ obtained through aggregation processing by W_{filter} to filter out anomalous features at the edges of different semantic categories, resulting in the RGB modality feature $F_{\text{filter}}^{\text{rgb}} \in \mathbb{R}^{C\times H\times W}$ and the depth modality feature $F_{\text{filter}}^{\text{depth}} \in \mathbb{R}^{C\times H\times W}$

$$F_{\text{filter}}^{\text{rgb}} = W_{\text{filter}} \otimes F_{\text{aggregated}}^{\text{rgb}} \tag{13}$$

$$F_{\text{filter}}^{\text{depth}} = W_{\text{filter}} \otimes F_{\text{aggregated}}^{\text{depth}} \tag{14}$$

where $\otimes$ is used for channelwise multiplication.

Finally, we combine the separated features with the aggregated features to obtain the calibrated and enhanced multimodal features

$$F_{\text{out}}^{\text{rgb}} = F_{\text{filter}}^{\text{rgb}} + F_{\text{enhanced}}^{\text{depth}} \tag{15}$$

$$F_{\text{out}}^{\text{depth}} = F_{\text{filter}}^{\text{depth}} + F_{\text{enhanced}}^{\text{rgb}} \tag{16}$$

where $F_{\text{out}}^{\text{rgb}}$ and $F_{\text{out}}^{\text{depth}} \in \mathbb{R}^{C\times H\times W}$ represent the RGB modality features and depth modality features enhanced through guided feature augmentation. The underlying concept of this operation is to leverage the complementary nature of spatial texture information and spatial geometric information of ground objects in outdoor scenes. By treating the complementary information as an offset, it refines and corrects the semantic information of classes under different modalities. The advantage of directly summing the elements is that this operation does not introduce any additional parameters and requires only a single tensor addition. Compared with concatenation or weighted fusion methods, this significantly reduces the computational load, providing lower computational latency for subsequent deployment of the model on drones, thereby enhancing its practicality.

Moreover, the features of RGB images and depth maps possess a natural advantage of being inherently calibrated. The depth maps in the CUS-Depth dataset are rendered from 3-D models using specific parameters, and their intrinsic and extrinsic parameters are aligned with those of the RGB images. Reorganizing channels for concatenated features may disrupt spatial correlation, and spatial transformations in weighted fusion (such as spatial transformers) can increase alignment errors. The addition operation directly fuses at the pixel level without altering the spatial dimensions of the feature maps, thus avoiding boundary blurring issues caused by interpolation or deformation operations.

This process makes the features of ground objects within the same semantic category more similar and those of ground objects in different semantic categories more distinct, thereby forming more robust class semantic information for subsequent feature encoding learning by the GOFEM.

C. Overview of GOFEM

As illustrated in Fig. 1, the feature encoder of SkyMamba consists of four layers of GOFEMs with different scales, designed to encode multiscale RGB-D features of ground objects in RS scenes. Between each layer of GOFEMs, a convolution with a 3×3 kernel and a stride of 2 is employed to downsample the RGB-D features.

In SkyMamba, the GOFEMs for each layer integrate a multimodal transformer (MMT) module and an RS spatial global attention (RSSGA) module. MMT module enhances cross-modal information exchange between RGB and depth data by leveraging the Transformer, and it employs a large kernel convolution operation to capture spatial geometric cues from the depth modality, thereby refining object edge details. The RSSGA models contextual information within RS scenes by utilizing a SSM from eight directions, effectively capturing global features across extensive spatial domains. This approach significantly enhances the correlation between the semantic information of objects within RS scenes. Finally, the cross-modal fusion features, geometric deep features, and RS spatial global features are concatenated and linearly projected to enhance the feature representation within the scene, thereby achieving feature updating. The detailed information of the GOFEM is shown in Fig. 3.

D. Multimodal Transformer Module

In RS scenes, RGB and depth modalities provide visual texture information and spatial geometric information of objects, respectively. To facilitate the interaction of information from these two modalities and enhance semantic category features, we propose the MMT module for multimodal feature interaction. The MMT module computes key and value vectors from the input RGB modality features $F_{\text{in}}^{\text{rgb}}$

$$V_{\text{value}} = LN\left(F_{\text{in}}^{\text{rgb}}\right) \tag{17}$$

$$V_{\text{key}} = LN\left(F_{\text{in}}^{\text{rgb}}\right) \tag{18}$$

where $F_{\text{in}}^{\text{rgb}}$ represents the RGB features input to the MMT module, *LN* denotes the MLP network operation, V_{value} and V_{key} represent the value and key vectors computed from the RGB features, respectively. During the computation of the query vector, the MMT module concatenates the input RGB and depth features and performs a pooling operation with a fixed size to obtain a query vector that integrates multimodal information

$$V_{\text{query}} = LN\left(GP_{k\times k}\left(\left[F_{\text{in}}^{\text{rgb}}, F_{\text{in}}^{\text{depth}}\right]\right)\right) \tag{19}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:20:32 UTC from IEEE Xplore. Restrictions apply.

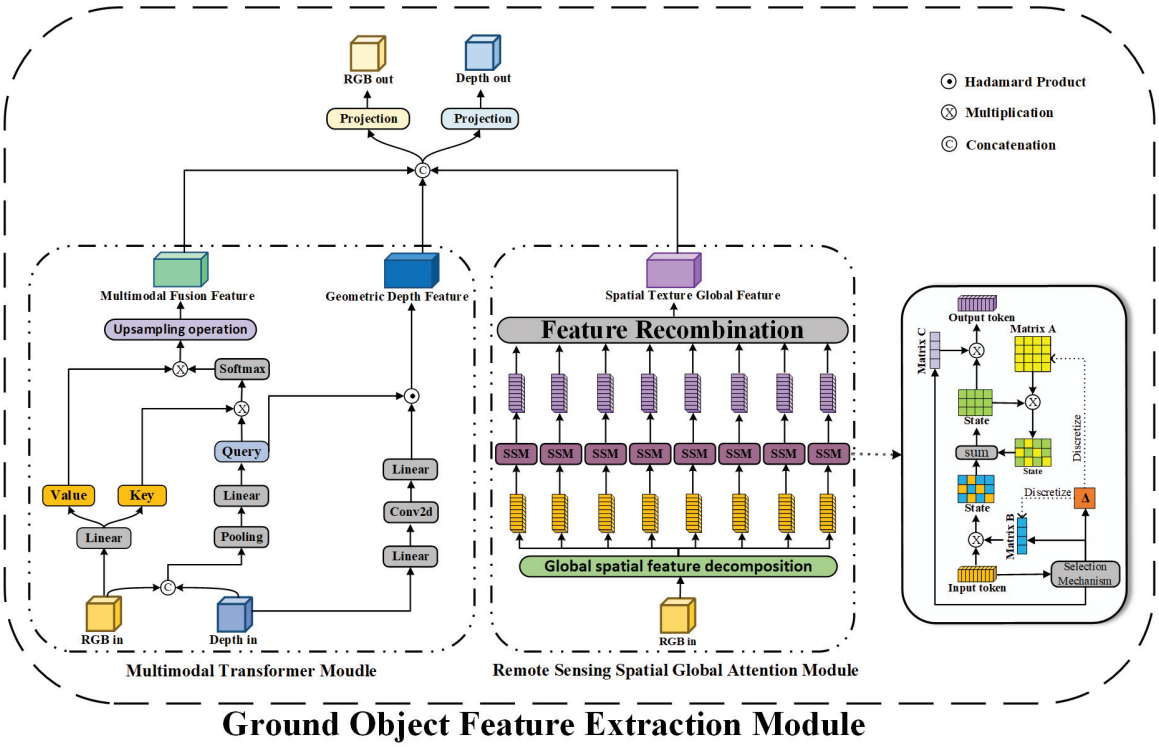


Fig. 3. Schematic of the ground scene feature extraction module. (a) Schematic of the MMT module, which employs a Transformer to integrate RGB and depth modality features, establishing cross-modal feature interaction relationships. This enhances 3-D perception in RS space, thereby further capturing semantic category information. (b) Schematic of the RSSGA module, which decomposes large-scale features in RS space into local features through global spatial decomposition. These local features are then reconstructed into feature sequences from eight directions. The SSM is employed to capture contextual information from different directions, and the modeling information from all eight directions is fused to obtain the global features of the entire RS space. (c) State-space attention calculation within the RSSGA module.

where $F_{\text{in}}^{\text{rgb}}$ and $F_{\text{in}}^{\text{depth}}$ represent the RGB and depth features input to the MMT module, respectively, $[\cdot,\cdot]$ denotes the cross-modal feature fusion operation, $GP_{k\times k}$ represents the adaptive average pooling operation with a fixed size of $k \times k$, and LN denotes the MLP network operation.

Following the acquisition of the $V_{\text{value}} \in \mathbb{R}^{c_i\times h\times w}$, $V_{\text{key}} \in \mathbb{R}^{c_i\times h\times w}$, and $V_{\text{query}} \in \mathbb{R}^{c_i\times k\times k}$, self-attention operation is conducted, and the result is upscaled to the appropriate dimension to derive the multimodal fusion feature $F_{\text{multimodal}} \in \mathbb{R}^{C\times H\times W}$

$$F_{\text{multimodal}} = UP\left(\text{Softmax}\left(\frac{V_{\text{query}}V_{\text{key}}^{T}}{\sqrt{c_i}}\right)V_{\text{value}}\right) \tag{20}$$

where UP denotes the bilinear upsampling operation, Softmax represents the softmax operation, c_i indicates the dimension of the current feature, and h and w denote the height and width of the current feature map.

Deep modalities in the RS space can provide more enhanced detail pertaining to object boundaries. To further preserve spatial structural information, we additionally employ large kernel convolutions within the MMT module to process depth features, and employ the Hadamard product in conjunction with the query vector to embed more information, thereby augmenting the geometric structural details of objects within the RS domain

$$F_{\text{geometric}} = LN\left(\text{DConv}\left(LN\left(\mathcal{F}_{\text{in}}^{\text{depth}}\right)\right)\right) \odot V_{\text{query}} \tag{21}$$

where $F_{\text{geometric}} \in \mathbb{R}^{C\times H\times W}$ represents the depth features processed by large kernel depthwise convolution and subsequently multiplied with the query vector, $F_{\text{in}}^{\text{depth}}$ denotes the depth features input into the MMT module, LN stands for the MLP network operation, DConv signifies the large kernel depthwise convolution operation, $\odot$ denotes the Hadamard product operation, and V_{query} represents the query vector computed from the RGB and depth modalities.

E. RS Spatial Global Attention Module

In RS images acquired from a UAV perspective, the spatial distribution of features significantly differs from that observed from ground-level road views. Objects within such images exhibit strong correlations and rich contextual information, with feature distributions potentially oriented in any direction. To effectively capture the multidirectional global information inherent in RS spaces, we propose the RSSGA module. This module decomposes RS spatial features and reconstructs them into feature sequences along eight distinct directions. The SSM is employed to capture contextual information within each feature sequence. Finally, the modeled features from all directions are integrated to obtain the global features within the RS space.

We decompose the RS images input into the RSSGA module into 12 local image patches for global spatial feature separation. Subsequently, the decomposed local patches are reconstructed into feature sequences along horizontal, vertical,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:20:32 UTC from IEEE Xplore. Restrictions apply.

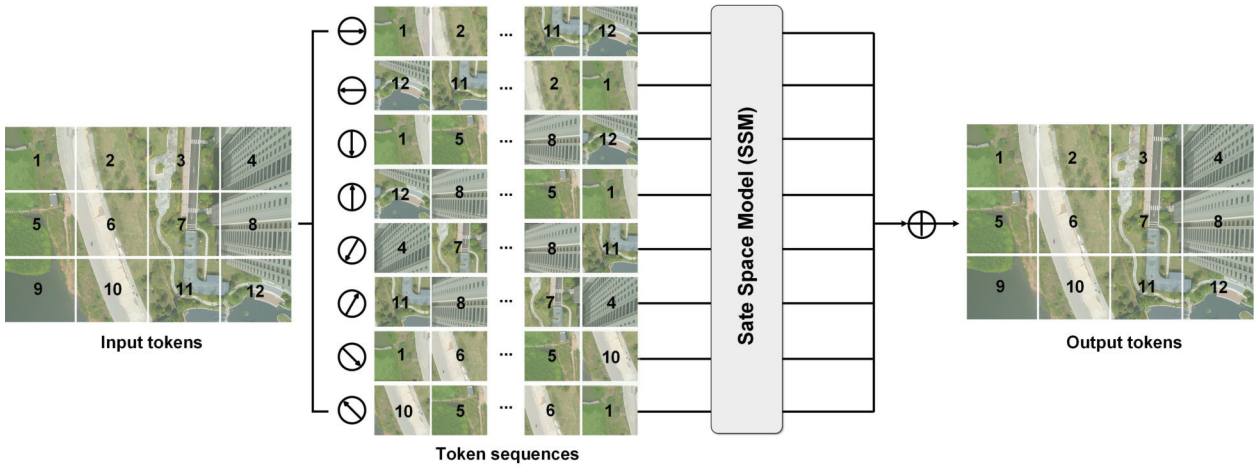


Fig. 4. Schematic of the process for global spatial feature modeling in RS images. High-resolution RS images are separated and reconstructed into sequences along specific directions. Contextual information is modeled in each direction using a SSM, and finally, contextual information from all directions is fused to achieve global spatial feature modeling in RS.

diagonal, and anti-diagonal directions, as shown in Fig. 4. The feature sequences reconstructed in different directions facilitate the encoder in extracting richer global spatial features from RS images.

The feature sequences reconstructed from eight different directions are modeled globally using SSM along a new dimension. The SSM can encapsulate the state information within dynamic systems, thus effectively modeling contextual information of features. To dynamically perceive and reason about contextual information within RS space, the RSSGA module first adaptively obtains the corresponding state transition matrix A_i, input state matrix B_i, and output state matrix C_i for the ith local feature map F_i of each input directional feature sequence. The matrices A_i and B_i are discretized using the zero-order hold technique with parameter Δ

$$\bar{A}_i = e^{\Delta A_i} \tag{22}$$

$$\bar{B}_i = \left(e^{\Delta A_i} - I\right) A_i^{-1} B_i \tag{23}$$

where $\bar{A}_i$ and $\bar{B}_i$ represent the discretized state transition matrix and input state matrix, respectively, Δ denotes the discretization parameter, and I denotes the identity matrix.

Subsequently, the state matrix, which dynamically varies based on the adaptive parameter Δ, selectively focuses on contextual information within different local feature maps. The current state representation State_i^1 is obtained by multiplying the input local feature F_i with the input state matrix $\bar{B}_i$

$$\text{State}_i^1 = \bar{B}_i F_i. \tag{24}$$

The current local feature state is combined with the retained state $\text{State}_{i-1}^{\text{reserved}}$ of preceding local features to model contextual information. The retained state $\text{State}_{i-1}^{\text{reserved}}$ of preceding local features selectively preserves important spatial characteristics to be assigned to the current local feature state, resulting in a new feature state State_i^2

$$\text{State}_i^2 = \text{State}_i^1 + \text{State}_{i-1}^{\text{reserved}} \tag{25}$$

where the retained state $\text{State}_{i-1}^{\text{reserved}}$ of preceding local features is obtained by multiplying the corresponding state transition matrix $\bar{A}_{i-1}$ with the state State_{i-1}^2 of the preceding local features

$$\text{State}_{i-1}^{\text{reserved}} = \bar{A}_{i-1}\text{State}_{i-1}^2. \tag{26}$$

Finally, the new local feature state State_i^2 is multiplied with the output state matrix C_i to yield the new local spatial feature F_i'

$$F_i' = C_i \text{State}_i^2. \tag{27}$$

The new local spatial feature integrates the spatial distribution information of preceding features, enabling the modeling of contextual information in specific directions. Meanwhile, the current local feature state State_i^2 is multiplied by the corresponding state transition matrix $\bar{A}_i$ to obtain the retained state $\text{State}_i^{\text{reserved}}$, which provides preceding feature information for the input features in the next stage. Following the reconstruction of feature sequences along various distinct directions, the integration and recombination of all local feature sequences facilitate the comprehensive modeling of global features within the RS space

$$F_{\text{global}} = \text{Recombination}\left(\sum_{j=1}^{8}\left[F_1', \ldots, F_{12}'\right]_j\right) \tag{28}$$

where F_{global} represents the global spatial features in RS, Recombination denotes the multidirectional feature sequence recombination operation, and $[F_1', \ldots, F_{12}']_j$ signifies the reconstructed feature sequence in a single direction.

F. Segmentation Decoder

Following the encoder, we choose to utilize the lightweight Hamburger decoder from SegNext [41] for decoding multiscale RGB-D features. Considering the robust feature extraction capabilities of the SkyMamba encoder, a lightweight decoder is adequate to achieve superior performance.

IV. Experiments

A. Dataset

UAV aerial RS images provide higher resolution than satellite RS images, offering more detailed information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:20:32 UTC from IEEE Xplore. Restrictions apply.

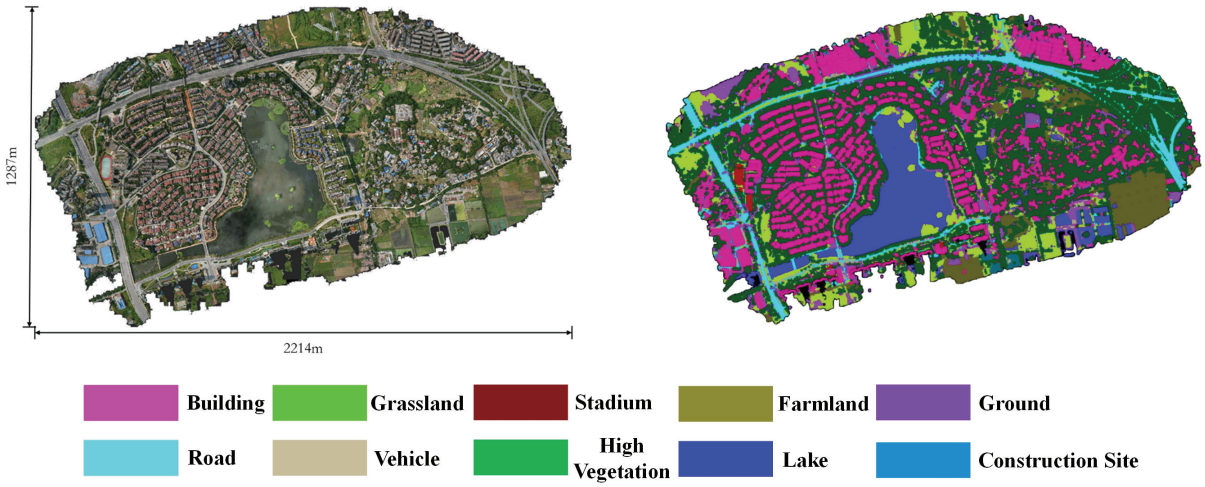


Fig. 5. Schematic illustration of the scenes covered by the CUS-Depth V1 dataset. (Left) Schematic representation of the areas covered by the original RGB images in the CUS-Depth V1 dataset; (Right) Schematic representation of the distribution of semantic labels corresponding to the RGB images in the CUS-Depth V1 dataset.

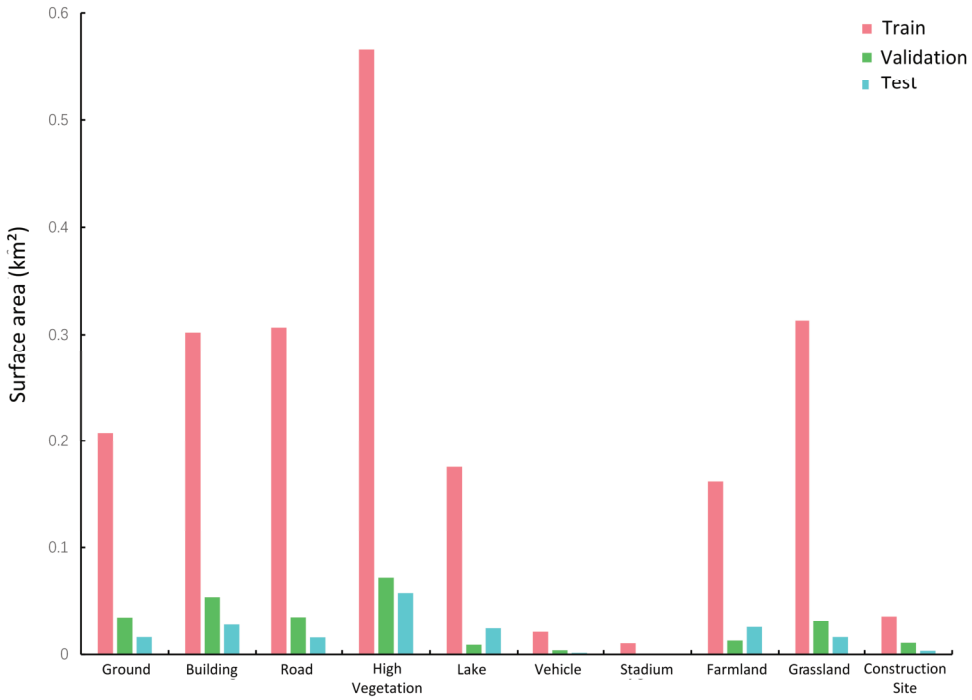


Fig. 6. Bar chart of imbalanced ground object semantic category distribution in large-scale scenes.

Furthermore, the incorporation of depth information can significantly enhance the performance of semantic segmentation in RS images. We introduce a novel benchmark dataset for UAV aerial RS RGB-D image semantic segmentation, named CUS-Depth V1, which is an extension of the recently proposed comprehensive 3-D semantic segmentation dataset CUS3D [59].

During the dataset construction process, we utilized a DJI M300 drone equipped with a SHARE 102s five-lens camera to capture large-scale aerial image sequences. Throughout the data acquisition process, we recorded the extrinsic parameters of each image to preserve its spatial information in the real-world coordinate system. Subsequently, we reconstructed high-precision 3-D models from the collected image sequences and performed semantic annotations based on these models, thereby constructing the CUS3D dataset [59]. Building upon this foundation, we further employed volumetric rendering integration techniques and leveraged the recorded image poses to compute the corresponding depth maps for each RGB image, generating RGB-D image pairs. This process ensures the physical consistency of the depth information within the CUS-Depth dataset, allowing the depth values to accurately reflect the geometric structure of the scene rather than relying on predictions from conventional depth estimation methods.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:20:32 UTC from IEEE Xplore. Restrictions apply.

TABLE I
HYPERPARAMETER SETTINGS FOR COMPARATIVE EXPERIMENTS OF EACH BASELINE METHOD ON THE CUS-DEPTH V1 AND CUS-DEPTH V2 DATASETS

Methods	Hyperparameter Setting					
	Epoch	Base Learning Rate	Batch Size	Momentum	Optimizer	Crop size
SkyMamba	150	1e-3	72	0.9	AdamW	512×512
ACNet [34]	200	2e-3	40	0.9	AdamW	512×512
SA-Gate [35]	200	1e-3	34	0.9	AdamW	512×512
TokenFusion [61]	200	2e-3	50	0.9	AdamW	512×512
CMX [62]	150	1e-3	50	0.9	AdamW	512×512
CMNext [63]	150	6e-5	25	0.9	AdamW	512×512
DFormer [36]	200	6e-5	50	0.9	AdamW	512×512
RS3Mamba [26]	150	1e-2	40	0.9	AdamW	512×512
RS-Mamba [14]	150	1e-3	30	0.9	AdamW	512×512
SegFormer [64]	100	1e-4	80	0.9	AdamW	512×512

The CUS-Depth V1 dataset comprises a total of 4336 orthographic and 45° oblique-view UAV aerial RGB images with corresponding depth maps from Changsha, China. The images have a resolution of 6144×4096 pixels. Among these, 2601 images are designated for training, 867 for validation, and 868 for testing. The CUS-Depth V1 dataset encompasses a variety of outdoor scenes, including urban, suburban, and rural areas, covering an area of 2.85 km^2. As shown in Fig. 5, the dataset provides ten common outdoor semantic categories, namely, building, road, high vegetation, grassland, lake, vehicle, farmland, construction site, stadium, and ground. In addition, as illustrated in Fig. 6, we present a bar chart that visually depicts the distribution of different categories across large-scale scenes.

To further promote the development of research on semantic segmentation of UAV aerial RS RGB-D images, we introduce the CUS-Depth V2 dataset, which presents increased challenges. This dataset maintains the same image resolution, quantity, and scene coverage as the basic version, CUS-Depth V1. However, CUS-Depth V2 offers a richer set of semantic categories, including building, villa, vehicle, pedestrian, building material, nonmotorized vehicle, grassland, high vegetation, road, footpath, farmland, dirt ground, lake, athletic field, basketball court, children's playground, power facility, and sundries, totaling 18 distinct semantic categories.

B. Benchmark Methods and Experimental Details

To assess the efficacy of SkyMamba in the semantic segmentation of UAV aerial RS RGB-D images, we conducted comparative experiments on the CUS-Depth V1 and CUS-Depth V2 datasets using six representative RGB-D image semantic segmentation methods: ACNet [34], SA-Gate [35], TokenFusion [60], cross modal fusion for RGB-X [61], CMNext [62], and DFormer [36], as well as two RS semantic segmentation baseline methods: RS3Mamba [26] and RS-Mamba [14], and one classic semantic segmentation model SegFormer [63].

1) Data Preprocessing: To mitigate potential errors in performance evaluation arising from sophisticated data augmentation techniques, we utilized straightforward augmentation methods, including horizontal flipping and random scaling. For the sake of research expediency, we segmented the input RS images into nonoverlapping 512×512 pixel blocks for model training. All baseline methods tested on the same dataset employed identical data preprocessing strategies to maintain experimental consistency.

2) Experimental Setup: We deployed SkyMamba and other baseline comparative experiments on a single NVIDIA Tesla A800 GPU using the PyTorch framework [64]. The batch size was adjusted according to the specific requirements of each model to maximize computational efficiency. We integrated the cross-entropy loss and Dice coefficient to construct the loss function for SkyMamba to optimize model performance. The AdamW optimizer was utilized with an initial learning rate of 0.001 and a weight decay of 0.001 to update the model parameters, which were trained for 150 epochs to ensure convergence. Table I presents the comprehensive experimental parameter settings for SkyMamba and other benchmark methods across the CUS-Depth V1 and CUS-Depth V2 datasets.

3) Evaluation Metrics: To evaluate the performance of the semantic segmentation model, we utilized intersection over Union (IoU) for each semantic class, mIoU, and mean Accuracy (mAcc) as evaluation metrics. The IoU and mIoU metrics assess the overlap between the segmentation results and the ground truth, thereby evaluating the spatial accuracy of the model's predictions. The mAcc metric measures the number of correctly classified pixels in the segmentation results, reflecting the overall performance of the model in pixel classification.

C. Comparative Experimental Results and Discussion

To validate the advanced performance of the proposed SkyMamba on semantic segmentation of UAV aerial RS RGB-D images, we compared the final SkyMamba model with baseline RGB-D semantic segmentation methods and RS semantic segmentation baseline methods on the CUS-Depth V1 and CUS-Depth V2 datasets. Table II presents the experimental results of various semantic segmentation methods on the CUS-Depth V1 dataset. Our SkyMamba achieved the highest mIoU (78.46%) on the CUS-Depth V1 dataset, which is 4.41% higher than the latest RGB-D semantic segmentation

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:20:32 UTC from IEEE Xplore. Restrictions apply.

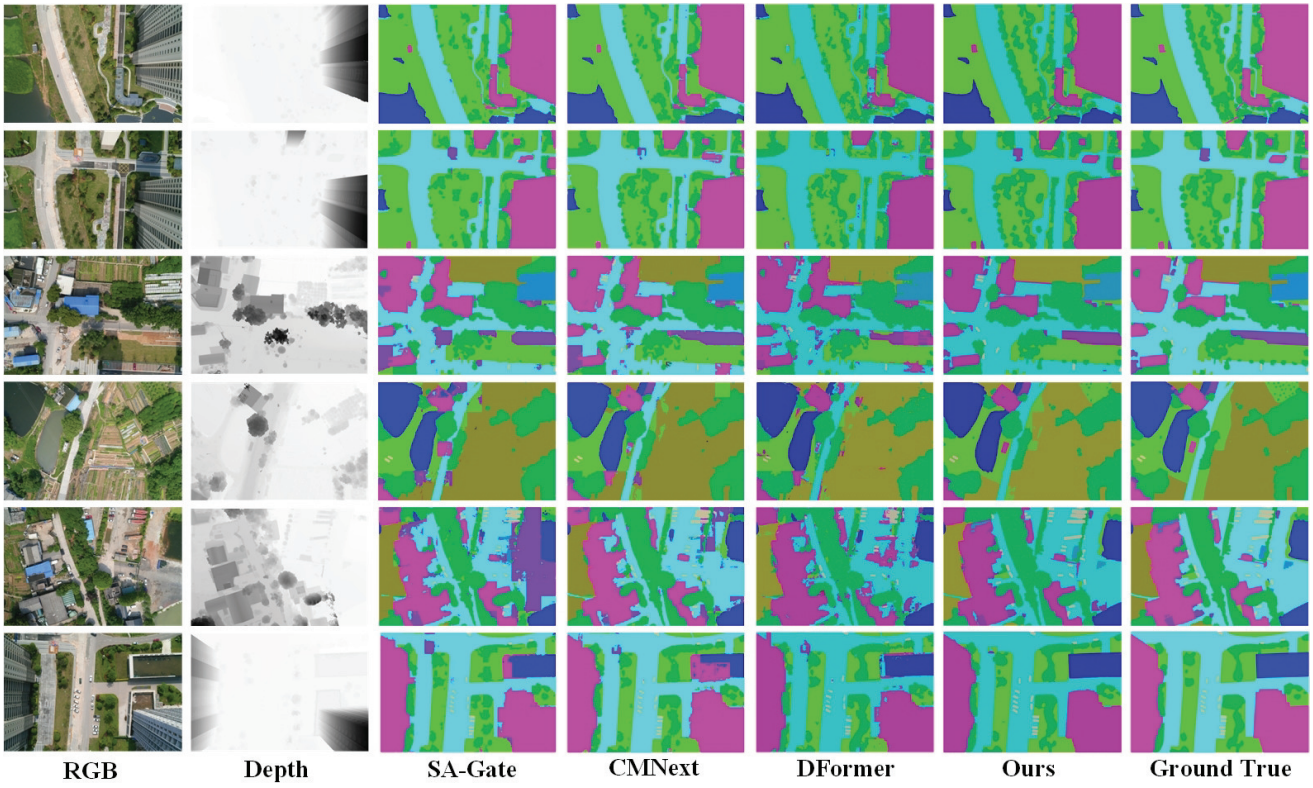


Fig. 7. Qualitative visual comparison of SkyMamba (Ours), SA-Gate [35], CMNext [62], and DFormer [36] on the CUS-Depth V1 dataset.

TABLE II

COMPARISON RESULTS OF SKYMAMBA AND CONTRASTIVE METHODS ON THE CUS-DEPTH V1 DATASET. MEAN CLASS ACCURACY (MACC, %), MEAN IOU (MIOU, %), AND PER-CLASS IOU (%) ARE REPORTED. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

Class	Methods									
	SkyMamba	ACNet [34]	SA-Gate [35]	TokenFusion [61]	CMX [62]	CMNext [63]	DFormer [36]	RS-Mamba [14]	RS3Mamba [26]	SegFormer [64]
Building	**88.80**	85.31	85.74	87.29	87.39	86.18	87.52	85.61	82.54	78.42
Road	**83.66**	75.81	79.41	80.82	81.09	80.08	80.86	79.58	77.50	71.80
Vehicle	**81.45**	64.74	66.25	76.35	70.61	68.96	75.82	68.26	62.78	36.48
Grassland	**66.54**	62.48	61.94	62.48	63.05	63.83	61.45	61.07	48.31	50.69
High Vegetation	85.28	**89.38**	83.38	84.92	85.31	84.65	84.95	84.19	82.25	78.70
Stadium	90.47	81.60	87.98	88.45	86.46	**91.18**	88.92	83.03	73.12	80.45
Lake	95.87	87.07	95.19	94.44	**96.41**	95.29	95.22	94.37	87.80	92.96
Farmland	**77.14**	69.80	71.19	75.10	76.04	73.61	74.71	71.94	67.13	61.17
Construction Site	**64.07**	36.95	55.33	55.15	48.73	50.21	51.95	39.44	31.29	0.06
Ground	**51.28**	30.74	43.38	44.90	44.25	42.54	39.13	36.52	31.58	21.70
mIoU (%)	**78.46**	68.39	72.98	71.96	73.93	73.65	74.05	70.41	64.43	52.04
mAcc (%)	87.95	81.87	**88.68**	81.81	83.80	81.93	83.40	78.21	75.71	59.46

method, DFormer [36] (74.05%). In addition, SkyMamba also achieved the second-highest mAcc (87.95%) on the CUS-Depth V1 dataset, second only to SA-Gate [35] (88.68%). Across various semantic categories, SkyMamba achieved the highest performance in the majority of categories in terms of IoU metrics. It ranked second only to CMX [61] (96.41%) with an IoU of 95.87% in the lake category. However, it significantly outperformed other methods with an IoU of 81.45% in the vehicle category, which is a small class with fewer instances. We present some visual inference results of SkyMamba and several comparative methods on the test set of the CUS-Depth V1 dataset in Fig. 7.

In practical scenarios, RS scenes often contain numerous fine and intricately distributed objects. Preliminary experiments on the CUS-Depth V1 dataset indicate that SkyMamba exhibits superior detail-capturing capabilities compared with other methods in large-scale spatial contexts. To further validate the semantic segmentation abilities of SkyMamba in such expansive spaces, comparative experiments were conducted on the more challenging CUS-Depth V2 dataset. Table III presents the experimental results of various semantic segmentation methods on the CUS-Depth V2 dataset. The results indicate that SkyMamba achieved the highest mIoU (63.82%) even when faced with a complex RS scene

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:20:32 UTC from IEEE Xplore. Restrictions apply.

TABLE III

COMPARISON RESULTS OF SKYMAMBA AND CONTRASTIVE METHODS ON THE CUS-DEPTH V2 DATASET. MEAN CLASS ACCURACY (MACC, %), MEAN IOU (MIOU, %), AND PER-CLASS IOU (%) ARE REPORTED. THE BEST VALUES ARE HIGHLIGHTED IN BOLD

Class	Methods									
	SkyMamba	ACNet [34]	SA-Gate [35]	TokenFusion [61]	CMX [62]	CMNext [63]	DFormer [36]	RS-Mamba [14]	RS3Mamba [26]	SegFormer [64]
Building	81.77	74.77	78.33	81.97	80.27	80.89	**82.30**	81.33	60.65	61.04
Villa	77.56	46.92	74.53	**79.29**	74.95	78.22	77.89	78.58	23.59	60.37
Building material	56.19	**67.58**	43.29	54.48	45.28	49.44	53.95	46.22	0.0	0.05
Vehicle	78.67	54.96	65.07	78.32	67.87	65.94	**79.23**	71.75	31.19	32.62
Pedestrian	**7.95**	0.0	0.0	2.68	1.12	1.15	3.28	0.0	0.0	0.0
Non-motorized vehicle	18.96	0.0	0.0	**24.33**	13.73	10.58	22.06	0.0	0.0	0.0
Grassland	**65.15**	61.26	58.61	62.09	61.65	63.72	59.33	62.37	41.21	46.58
High Vegetation	**85.21**	76.43	82.11	85.01	84.48	84.47	83.17	85.01	80.01	77.47
Road	**81.33**	45.97	75.17	77.94	78.76	77.84	78.81	79.50	69.13	66.06
Footpath	60.99	**69.58**	48.59	53.16	52.15	52.87	52.03	52.34	3.08	16.67
Farmland	74.05	34.08	69.50	**74.33**	72.95	73.52	72.47	72.65	23.78	56.24
Dirt Ground	**47.90**	41.86	35.84	37.11	40.87	43.12	39.82	41.73	4.15	17.72
Lake	95.57	92.60	94.04	94.73	**96.11**	95.29	95.06	95.34	75.91	92.32
Athletic Field	**97.45**	81.37	92.20	92.38	93.44	94.87	94.71	94.78	44.07	81.35
Basketball Court	**91.56**	31.02	82.69	83.94	87.52	90.61	81.67	85.26	0.03	73.55
Children's playground	**62.32**	37.04	44.28	36.38	58.29	56.55	42.89	0.0	0.0	0.0
Power Facility	42.93	19.83	29.93	39.28	46.89	**55.17**	30.21	0.0	0.0	0.0
Sundries	**23.19**	0.0	22.46	19.47	14.08	16.03	14.13	8.86	0.0	0.0
mIoU (%)	**63.82**	46.40	55.37	59.83	59.47	60.57	59.06	53.10	25.38	35.90
mAcc (%)	72.00	64.57	**75.44**	69.85	67.85	68.96	67.46	59.61	31.68	42.95

with 18 semantic categories. This represented an improvement of 3.25% over the second-highest method, CMNext [62] (60.57%), and a 4.76% increase compared with the latest RGB-D semantic segmentation method, DFormer [36] (59.06%). In addition, SkyMamba attained the second-highest mAcc (72%). Notably, SkyMamba demonstrates superior segmentation performance on fine-grained categories with fewer distributions, such as vehicle, pedestrian, athletic field, children's playground, and sundries. It also demonstrates a more equitable semantic segmentation performance on complex objects within large-scale spaces, effectively addressing the challenge of imbalanced semantic category distributions in the dataset to a certain extent. We present some visual inference results of SkyMamba and several comparative methods on the test set of the CUS-Depth V2 dataset in Fig. 8.

Across both the CUS-Depth V1 and V2 datasets, SkyMamba demonstrates a significant performance advantage over the RS-specific baseline methods, RS-Mamba [14] and RS3Mamba [26]. On CUS-Depth V1, SkyMamba achieves a mIoU of 78.46%, surpassing RS-Mamba (70.41%) and RS3Mamba (64.43%) by 8.05 and 14.03 percentage points, respectively. In the more complex CUS-Depth V2 dataset, this advantage becomes even more pronounced, with mIoU differences reaching 10.72% and 38.44%. At the category level, SkyMamba consistently outperforms in challenging environments, such as "construction site" in CUS-Depth V1, and in long-tail small-object segmentation, exemplified by "nonmotorized vehicle" in CUS-Depth V2. These results validate the effectiveness of SkyMamba's global state-space modeling in capturing large-scale continuous structures and preserving the integrity of small objects, highlighting its superiority and robustness in multiscale semantic segmentation tasks for RS.

In comparative experiments with the classic semantic segmentation model SegFormer [63], SkyMamba achieved an mIoU of 78.46% on the CUS-Depth V1 dataset, while SegFormer [63] only reached 52.04%. On the CUS-Depth V2 dataset, SkyMamba attained an mIoU of 63.82%, compared with SegFormer [63]'s 35.90%. This disparity arises because traditional semantic segmentation models like SegFormer [63], which operate solely on RGB data, are unable to leverage the spatial structural information that is crucial for RS scenes. Depth information provides insights into spatial relationships between objects, geometric structures, and absolute/relative scales, which are particularly important in shadowed areas, low-texture regions (such as water bodies, roads, and large vegetation), and scenes with overlapping structures (such as dense urban environments). The segmentation capabilities of SegFormer [63], which rely exclusively on RGB texture and color, are significantly limited when faced with the complex factors commonly found in the CUS-Depth dataset, such as dense shadows, extensive low-texture areas, and structural ambiguities caused by perspective or occlusion. The improvement in semantic segmentation metrics demonstrated by SkyMamba clearly illustrates the effectiveness of integrating depth information, which provides essential spatial structure and geometric cues, effectively addressing the challenges faced by pure RGB models like SegFormer in distinguishing complex shadowed areas and low-texture land cover.

The CUS-Depth V2 dataset, compared with the CUS-Depth V1 dataset, includes a greater variety of object categories and more detailed object information. This fine-grained object feature set presents a greater challenge for semantic segmentation research. The fundamental reason for the poor segmentation performance and blurred boundaries of certain object categories, such as pedestrians and nonmotorized vehicles, is that these categories occupy a significantly smaller pixel area in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:20:32 UTC from IEEE Xplore. Restrictions apply.

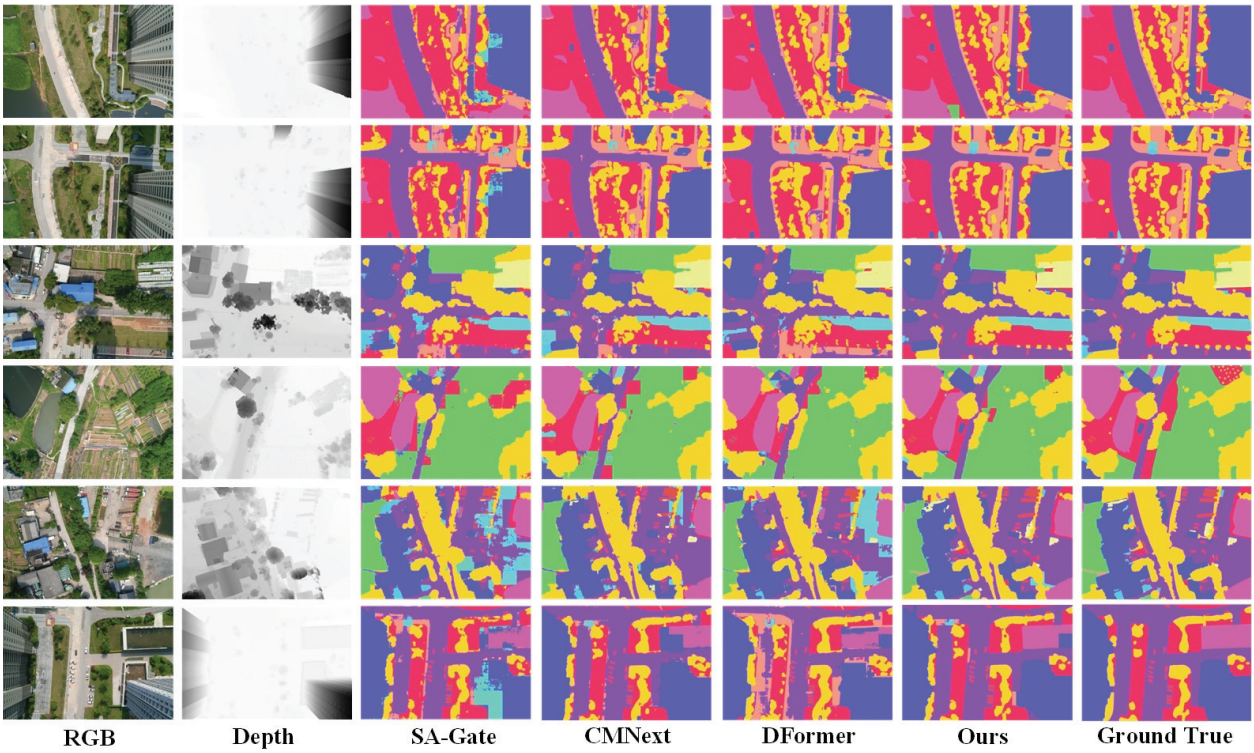


Fig. 8. Qualitative visual comparison of SkyMamba (Ours), SA-Gate [35], CMNext [62], and DFormer [36] on the CUS-Depth V2 dataset.

images compared to larger objects like roads and buildings. This disparity makes it difficult for models to learn their intricate features. In addition, small categories often appear infrequently in the training set due to the presence of scenes without pedestrians or nonmotorized vehicles, resulting in a lack of sufficient positive samples to support robust feature learning. During the feature extraction process through convolutional downsampling, small sample objects may be reduced to just a few pixels, leading to a severe loss of texture and geometric information, which in turn hampers the reconstruction of details in high-level feature maps. Small sample objects are often closely adjacent to the background or similar objects, making it challenging for low-resolution feature maps to distinguish subtle boundaries, thereby significantly increasing the edge error rate. SkyMamba places greater emphasis on the global segmentation performance in low-altitude RS spaces. While it may lag behind other methods in the segmentation performance of certain small object categories, as shown in Table III, the SkyMamba model demonstrates significant improvements in the semantic segmentation performance of small object categories compared to baseline methods such as RS-Mamba [14] and RS3Mamba [26].

In summary, SkyMamba achieves state-of-the-art performance in the field of UAV aerial RGB-D image semantic segmentation. In RS scenarios, ground object features are distributed in diverse orientations and cover large spatial scales. The architecture of SkyMamba effectively integrates visual texture information and geometric depth information while capturing global context and substantive spatial semantic category information from various directions across large-scale spaces. This capability enables highly accurate semantic segmentation of RS images. Compared to the latest RGB-D semantic segmentation baseline DFormer [36], SkyMamba demonstrates superior segmentation precision in most categories, particularly in complex scenes, as shown in Fig. 9. In addition, compared to RS-Mamba [14] and RS3Mamba [26], SkyMamba significantly improves segmentation accuracy across various categories, achieving the highest mIoU while maintaining a competitive mAcc. Although some methods surpass SkyMamba in specific categories, its overall segmentation performance remains highly competitive. These results validate the effectiveness of SkyMamba's architectural design in handling challenging RS environments and its robustness in balancing fine-grained and large-scale segmentation tasks.

D. Ablation Experiments and Discussion

The SkyMamba model is composed of multiple GOFEMs, each incorporating an MMT module and an RSSGA module. In addition, a lightweight FGE module is designed for feature refinement prior to feature encoding. To further investigate the effectiveness of each component in the SkyMamba model on semantic segmentation performance, we conducted an ablation study on the CUS-Depth V1 dataset with identical hyperparameter settings.

1) Componentwise Ablation Study: We individually removed the FGE module, MMT module, and RSSGA module from SkyMamba and conducted experiments on the CUS-Depth V1 dataset. The results are presented in Table IV. The ablation study results demonstrate that all components are essential for SkyMamba. The FGE

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:20:32 UTC from IEEE Xplore. Restrictions apply.

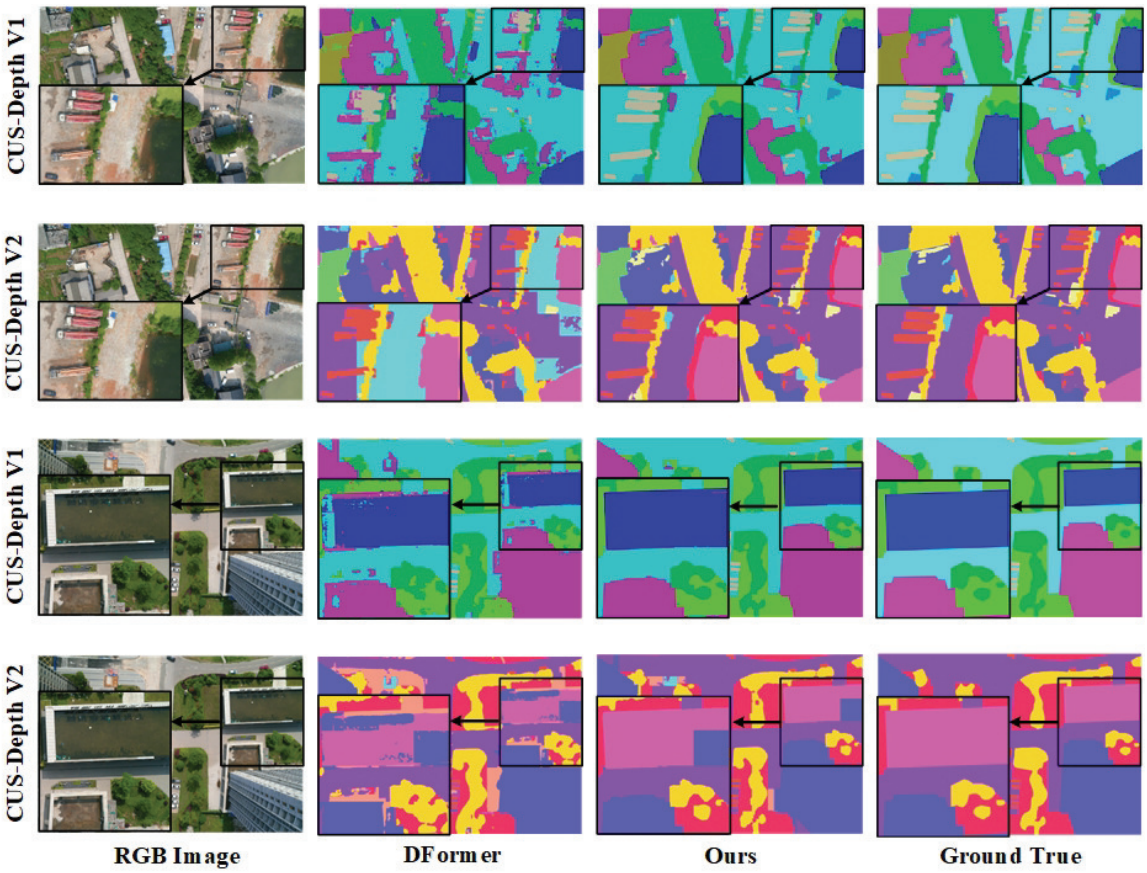


Fig. 9. Visualization details comparison between SkyMamba (Ours) and DFormer [36] on the CUS-Depth dataset. We can observe that SkyMamba (Ours) is capable of more accurately segmenting object shapes in complex scenes.

TABLE IV
ABLATION RESULTS ON THE COMPONENTS OF THE SKYMAMBA MODEL

Model	mIoU(%)	Params	FLOPs
SkyMamba	78.46	57.91M	68.83G
w/o FGE	76.18 (2.28 ↓)	57.90M	68.82G
w/o MMT	74.59 (3.87 ↓)	53.57M	65.89G
w/o RSSGA	72.47 (5.99 ↓)	55.38M	52.87G

TABLE V
ABLATION STUDY ON THE GOFEM MODULE

Model	SkyMamba	No1	No2	No3	No4
mIoU (%)	78.46	76.27	76.88	75.84	68.13

module improves the semantic segmentation performance of SkyMamba by 2.28% in mIoU, while introducing only a 0.01M increase in model parameters and a 0.01G increase in computational complexity. Given its minimal impact on the overall model in terms of parameter size and computational cost, the module can be considered negligible, demonstrating its lightweight advantage. The MMT module improves semantic segmentation performance by 3.87% in mIoU, effectively integrating multimodal features for efficient semantic segmentation. The RSSGA module provides the most significant performance boost to SkyMamba, increasing mIoU by 5.99%, attributable to its global spatial modeling capability that effectively captures and provides global contextual information for large-scale RS spaces.

2) Ablation Study on the Details of the GOFEM Module: To verify the effectiveness of the proposed multiscale feature encoder, we conducted an ablation study on the GOFEM module. The feature encoder consists of four layers of GOFEM modules at different scales. These modules are designed to encode multiscale RGB-D features of ground objects in RS scenes, capturing complementary information from objects of varying sizes and complexities. As shown in Table V, the experimental results indicate that removing any of these modules leads to a significant decline in segmentation performance. This confirms that each module plays a crucial role in encoding different features of RS data.

3) Ablation Study on the Details of the FGE Module: Within the FGE module, HFA and HFS operations are employed to suppress anomalous feature responses in the edge regions of objects within RS scenes, while enhancing homogeneous feature responses within the same class of objects. This process reinforces the critical semantic category information of objects in the spatial context. To further investigate the effectiveness of HFA and HFS operations within the module, we conducted ablation studies on both

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:20:32 UTC from IEEE Xplore. Restrictions apply.

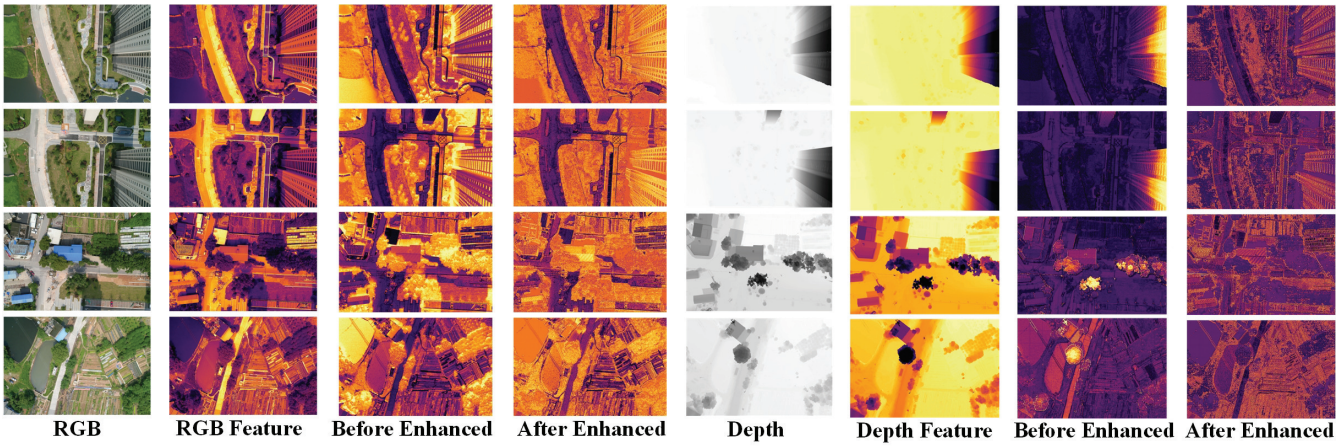


Fig. 10. Visualization of RGB and depth modality features processed by the FGE module and those not processed by the FGE module. These features are selected from the output of the first layer of the network.

TABLE VI

ABLATION RESULTS ON THE HFA AND HFS OPERATIONS OF THE FGE MODULE

Model	mIoU(%)	Params	FLOPs
SkyMamba	78.46	57.91M	68.83G
w/o FGE	76.18 (2.28 ↓)	57.90M	68.82G
w/o HFA	76.99 (1.47 ↓)	57.91M	68.82G
w/o HFS	77.65 (0.81 ↓)	57.90M	68.82G

TABLE VII

EXPERIMENTAL RESULTS ON THE IMPACT OF DEPTH MODAL DATA ON SEMANTIC SEGMENTATION

Data Modal	mIoU(%)	Params	FLOPs
RGB	76.68	56.65M	66.23G
RGB+Depth	78.46 (1.58↑)	57.91M	68.83G

TABLE VIII

EXPERIMENTAL RESULTS OF FIXED POOLING KERNEL SIZE "$k \times k$" FOR Q-VECTOR CALCULATION IN THE MMT MODULE

Kernel Size	mIoU(%)	Params	FLOPs
3×3	77.49(0.97↓)	57.91M	65.32G
5×5	77.92(0.54↓)	57.91M	66.57G
7×7	78.46	57.91M	68.83G
9×9	78.44(0.02↓)	57.91M	71.93G

operations. Table VI presents the detailed ablation study results for the FGE module. The experimental results demonstrate that both HFA and HFS operations can enhance the performance of semantic segmentation. Specifically, the HFA operation improves the mIoU metric by 1.47%, while the HFS operation increases the mIoU metric by 0.81%. Overall, the feature aggregation and separation operations within the FGE module can effectively adaptively control the multimodal data flow based on the diverse information provided by cross-modal data, thereby enhancing the feature responses of the same semantic category while filtering out anomalous responses at the category boundaries. In addition, we visualized and compared the cross-modal features processed with and without the FGE module after the first network layer. As illustrated in Fig. 10, the RGB and depth features processed by the FGE module capture more comprehensive detail information.

TABLE IX

EXPERIMENTAL RESULTS OF DIFFERENT SCANNING DIRECTION COMBINATIONS IN THE RSSGA MODULE FOR RS IMAGE SEQUENCES

Directional Combination	mIoU(%)
Vertical	75.45(3.01 ↓)
Horizontal	75.16(3.30 ↓)
Diagonal	76.05(2.41 ↓)
Vertical+Horizontal	76.61(1.85 ↓)
Vertical+Diagonal	77.50(0.96 ↓)
Vertical+Diagonal	77.21(1.25 ↓)
Omnibearing	78.46

4) Study on the Details of the MMT Module: The MMT module effectively integrates RGB and depth modalities through an attention mechanism, achieving efficient semantic segmentation. To investigate the impact of geometric depth information on semantic segmentation, we experimented with using only RGB modality data as input and conducted the study with the same hyperparameter settings. The omission of a purely depth-based comparison is due to the nonindependence of geometric depth information. The essence of a depth map is a type of geometric signal that lacks direct semantic information. Therefore, pure depth information cannot independently accomplish multiclass semantic segmentation. As shown in Table VII, the experimental results indicate that compared to using only the RGB modality as input, the integration of geometric depth information through multimodal data input can improve the mIoU metric by 1.58%.

In the large-scale RS space, multimodal data that integrates visual texture information and geometric depth information

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:20:32 UTC from IEEE Xplore. Restrictions apply.

TABLE X
INFERENCE TIME (MILLISECONDS), MODEL PARAMETERS AND COMPUTATIONAL COMPLEXITY OF A SINGLE IMAGE BY DIFFERENT BASELINE METHODS

Class	Methods									
	SkyMamba	ACNet [34]	SA-Gate [35]	TokenFusion [61]	CMX [62]	CMNext [63]	DFormer [36]	RS3Mamba [26]	RS-Mamba [14]	SegFormer [64]
Times	66.13ms	10.19ms	84.89ms	90.16ms	24.09ms	13.55ms	54.69ms	9.45ms	5.02ms	6.28ms
Params	57.91M	116.60M	110.85M	45.91M	66.57M	58.69M	39.02M	49.66M	40.73M	84.7M
FLOPs	68.83G	106.11G	165.12G	80.32G	57.07G	87.36G	56.01G	63.31G	39.02G	183.3G

can more effectively achieve semantic segmentation of ground objects. Meanwhile, during the computation of the query vector in the MMT module, we observed that the semantic segmentation performance of SkyMamba initially improves with the increase of the fixed pooling kernel size, and it reaches its optimal performance when the fixed pooling kernel size is 7×7. Table VIII presents the experimental results for different fixed pooling kernel sizes.

5) Study on the Details of the RSSGA Module: The RSSGA module generates multiple sets of image patch sequences by performing global scans of the input RS images from different directions, such as horizontal, vertical, and diagonal. It employs the SSM to capture the complex global information in RS space across various directions, thereby effectively conducting comprehensive contextual modeling. We conducted seven groups of comparative experiments with various directional combinations to investigate the impact of sequence combinations with different scanning directions on the final segmentation performance. The experimental results are presented in Table IX. The experimental results demonstrate that feature sequences from single-direction scans can achieve similar semantic segmentation performance. Combining feature sequences from any two directions leads to improved segmentation performance, and full-direction scanning achieves the best semantic segmentation performance. The combination of more scanning directions provides richer contextual information, which aids in constructing global features in RS space and achieving superior semantic segmentation performance.

6) Structural Analysis: To evaluate the impact of the placement of the MFGEM module before and after GOFEM on model performance, we conducted a comparative experiment. Under the same conditions, we tested two structural configurations: placing MFGEM before GOFEM and placing MFGEM after GOFEM, using mIoU as the primary evaluation metric. The experimental results show that when MFGEM is placed before GOFEM (the structure adopted in this article), the model achieves an mIoU of 78.46%. However, when MFGEM is placed after GOFEM, the mIoU drops to 74.59%. Based on the above experimental results and analysis, we conclude that placing MFGEM before GOFEM is the optimal choice.

7) Practical Deployment Analysis: To verify the engineering applicability of the proposed method, we further conducted a quantitative analysis of the model's computational complexity and deployment adaptability. Based on the experimental data presented earlier, the model has 57.91M parameters, and the floating-point operations (FLOPs) for a single 6144×4096 resolution RS image amount to 68.83G. The RK3588 control board, a mainstream computing hardware for drones [65], features a neural processing unit computational capability of six TOPS and supports FP16/FP32 mixed-precision inference. Theoretically, this translates to a floating-point computing power of six TFLOPs, which satisfies the requirements for engineering deployment and practical application. As shown in Table X, the measured inference time of the proposed model is 66.13 ms, achieving a speedup of 24.03 ms compared to six baseline RGB-D semantic segmentation methods (with the slowest method taking 90.16 ms). Although the inference speed of our model is slightly lower than that of RS-Mamba [14] and RS3Mamba [26], these two baseline methods rely solely on RGB modal data for semantic segmentation, without integrating depth modal information, resulting in significantly lower segmentation accuracy. In conclusion, the proposed method achieves the best balance between segmentation accuracy and engineering practicality.

V. CONCLUSION

In this article, we present a novel framework for semantic segmentation of RS RGB-D images captured from UAVs, incorporating multiple modules for ground object scene feature extraction. This framework effectively integrates the visual texture information from the RGB modality with the geometric depth information from the depth modality. By utilizing the long-sequence modeling capabilities of SSMs, it effectively constructs global context information within large-scale RS environments, thereby achieving high-performance semantic segmentation from a UAV perspective. In addition, we integrate a lightweight MFGEM at the forefront of the framework. This module enhances the quality of multimodal feature representation by leveraging the complementary characteristics of RGB and depth data, thereby improving the performance of the model during the feature encoding learning phase.

Furthermore, we present a novel benchmark dataset for the semantic segmentation of RGB-D images in UAV RS, which is divided into two versions based on coarse and fine semantic label annotation categories to provide semantic segmentation challenges of varying difficulties. SkyMamba achieves state-of-the-art performance in comparative experiments across two versions of datasets, demonstrating exceptional small-scale object semantic segmentation capabilities even in scenarios with imbalanced semantic category distributions. Detailed comparative experiments and ablation studies validate the effectiveness of the proposed method. In addition, the constructed semantic segmentation dataset of UAV RS RGB-D images, annotated with both coarse and fine-grained semantic labels, facilitates the continuous development of complex environmental object analysis in the field of UAV RS.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:20:32 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] H. Yao, R. Qin, and X. Chen, "Unmanned aerial vehicle for remote sensing applications—A review," *Remote Sens.*, vol. 11, no. 12, p. 1443, Jun. 2019.

[2] Z. Zhang and L. Zhu, "A review on unmanned aerial vehicle remote sensing: Platforms, sensors, data processing methods, and applications," *Drones*, vol. 7, no. 6, p. 398, Jun. 2023.

[3] T. Wellmann et al., "Remote sensing in urban planning: Contributions towards ecologically sound policies?," *Landscape Urban Planning*, vol. 204, Dec. 2020, Art. no. 103921.

[4] S. Asadzadeh, W. J. D. Oliveira, and C. R. D. Souza Filho, "UAV-based remote sensing for the petroleum industry and environmental monitoring: State-of-the-art and perspectives," *J. Petroleum Sci. Eng.*, vol. 208, Jan. 2022, Art. no. 109633.

[5] M. Weiss, F. Jacob, and G. Duveiller, "Remote sensing for agricultural applications: A meta-review," *Remote Sens. Environ.*, vol. 236, Jan. 2020, Art. no. 111402.

[6] Z. Zheng, Y. Zhong, J. Wang, A. Ma, and L. Zhang, "Building damage assessment for rapid disaster response with a deep object-based semantic change detection framework: From natural disasters to man-made disasters," *Remote Sens. Environ.*, vol. 265, Nov. 2021, Art. no. 112636.

[7] H. Yu et al., "Methods and datasets on semantic segmentation: A review," *Neurocomputing*, vol. 304, pp. 82–103, Aug. 2018.

[8] A. Abdollahi, B. Pradhan, N. Shukla, S. Chakraborty, and A. Alamri, "Deep learning approaches applied to remote sensing datasets for road extraction: A state-of-the-art review," *Remote Sens.*, vol. 12, no. 9, p. 1444, May 2020.

[9] N. Zang, Y. Cao, Y. Wang, B. Huang, L. Zhang, and P. T. Mathiopoulos, "Land-use mapping for high-spatial resolution remote sensing image via deep learning: A review," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 5372–5391, 2021.

[10] B. Neupane, T. Horanont, and J. Aryal, "Deep learning-based semantic segmentation of urban features in satellite images: A review and meta-analysis," *Remote Sens.*, vol. 13, no. 4, p. 808, Feb. 2021.

[11] K. Huang, B. Shi, X. Li, X. Li, S. Huang, and Y. Li, "Multi-modal sensor fusion for auto driving perception: A survey," 2022, *arXiv:2202.02703*.

[12] N. Marchal, C. Moraldo, H. Blum, R. Siegwart, C. Cadena, and A. Gawel, "Learning densities in feature space for reliable segmentation of indoor scenes," *IEEE Robot. Autom. Lett.*, vol. 5, no. 2, pp. 1032–1038, Apr. 2020.

[13] H. Wang, J. Wang, and L. Agapito, "Co-SLAM: Joint coordinate and sparse parametric encodings for neural real-time SLAM," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 13293–13302.

[14] S. Zhao, H. Chen, X. Zhang, P. Xiao, L. Bai, and W. Ouyang, "RS-mamba for large remote sensing image dense prediction," 2024, *arXiv:2404.02668*.

[15] X. Chen, L. Gao, M. Zhang, C. Chen, and S. Yan, "Spectral–spatial adversarial multidomain synthesis network for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5518716.

[16] X. Chen, M. Zhang, and Y. Liu, "Target detection with spectral graph contrast clustering assignment and spectral graph transformer in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516916.

[17] S. Lee, S.-J. Park, and K.-S. Hong, "RDFNet: RGB-D multi-level residual feature fusion for indoor semantic segmentation," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 4980–4989.

[18] P. Wang et al., "Understanding convolution for semantic segmentation," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Lake Tahoe, NV, USA, Mar. 2018, pp. 1451–1460.

[19] Z. Zhang, Z. Cui, C. Xu, Y. Yan, N. Sebe, and J. Yang, "Pattern-affinitive propagation across depth, surface normal and semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 4106–4115.

[20] Y. Chen, T. Mensink, and E. Gavves, "3D neighborhood convolution: Learning depth-aware features for RGB-D and RGB semantic segmentation," in *Proc. Int. Conf. 3D Vis. (3DV)*, Sep. 2019, pp. 173–182.

[21] Y. He, W.-C. Chiu, M. Keuper, and M. Fritz, "STD2P: RGBD semantic segmentation using spatio-temporal data-driven pooling," 2016, *arXiv:1604.02388*.

[22] Z. Li, Y. Gan, X. Liang, Y. Yu, H. Cheng, and L. Lin, "LSTM-CF: Unifying context modeling and fusion with LSTMs for RGB-D scene labeling," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2016, pp. 541–557.

[23] Y. Cheng, R. Cai, Z. Li, X. Zhao, and K. Huang, "Locality-sensitive deconvolution networks with gated fusion for RGB-D indoor semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 3029–3037.

[24] S.-W. Hung, S.-Y. Lo, and H.-M. Hang, "Incorporating luminance, depth and color information by a fusion-based network for semantic segmentation," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2019, pp. 2374–2378.

[25] C. Li, W. Zhou, X. Zhou, and W. Yan, "Semantic progressive guidance network for RGB-D mirror segmentation," *IEEE Signal Process. Lett.*, vol. 31, pp. 2780–2784, 2024.

[26] X. Ma, X. Zhang, and M.-O. Pun, "RS^{3}Mamba: Visual state space model for remote sensing image semantic segmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024, Art. no. 6011405.

[27] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision mamba: Efficient visual representation learning with bidirectional state space model," 2024, *arXiv:2401.09417*.

[28] Y. Liu et al., "VMamba: Visual state space model," 2024, *arXiv:2401.10166*.

[29] S. Kong and C. Fowlkes, "Recurrent scene parsing with perspective understanding in the loop," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 956–965.

[30] D. Lin, G. Chen, D. Cohen-Or, P.-A. Heng, and H. Huang, "Cascaded feature network for semantic segmentation of RGB-D images," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 1311–1319.

[31] X. Chen, K.-Y. Lin, C. Qian, G. Zeng, and H. Li, "3D sketch-aware semantic scene completion via semi-supervised structure prior," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4192–4201.

[32] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[33] J. Wang, Z. Wang, D. Tao, S. See, and G. Wang, "Learning common and specific features for RGB-D semantic segmentation with deconvolutional networks," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2016, pp. 664–679.

[34] X. Hu, K. Yang, L. Fei, and K. Wang, "ACNET: Attention based network to exploit complementary features for RGBD semantic segmentation," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2019, pp. 1440–1444.

[35] X. Chen et al., "Bi-directional cross-modality feature propagation with separation-and-aggregation gate for RGB-D semantic segmentation," in *Proc. 16th Eur. Conf. Comput. Vis. (ECCV)*, Aug. 2020, pp. 561–577.

[36] B. Yin, X. Zhang, Z. Li, L. Liu, M.-M. Cheng, and Q. Hou, "DFormer: Rethinking RGBD representation learning for semantic segmentation," 2023, *arXiv:2309.09668*.

[37] C. Papaioannidis, I. Mademlis, and I. Pitas, "Autonomous UAV safety by visual human crowd detection using multi-task deep neural networks," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, Jun. 2021, pp. 11074–11080.

[38] D. Alexiou et al., "Visual navigation based on deep semantic cues for real-time autonomous power line inspection," in *Proc. Int. Conf. Unmanned Aircr. Syst. (ICUAS)*, Jun. 2023, pp. 1262–1269.

[39] X. Xia, M. Koeva, and C. Persello, "Extracting cadastral boundaries from UAV images using fully convolutional networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 2455–2458.

[40] X. Xia, C. Persello, and M. Koeva, "Deep fully convolutional networks for cadastral boundary detection from UAV images," *Remote Sens.*, vol. 11, no. 14, p. 1725, Jul. 2019.

[41] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder–decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.

[42] Y. Zhang, C. Wu, T. Zhang, and Y. Zheng, "Full-scale feature aggregation and grouping feature reconstruction-based UAV image target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5621411.

[43] Y. Zhang, T. Liu, P. Yu, S. Wang, and R. Tao, "SFSANet: Multiscale object detection in remote sensing image based on semantic fusion and scale adaptability," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4406410.

[44] Y. Zhang, C. Wu, W. Guo, T. Zhang, and W. Li, "CFANet: Efficient detection of UAV image based on cross-layer feature aggregation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5608911.

[45] Y. Zhang, Y. Liu, W. Kang, and R. Tao, "VSS-net: Visual semantic self-mining network for video summarization," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 4, pp. 2775–2788, Apr. 2024.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:20:32 UTC from IEEE Xplore. Restrictions apply.

[46] Y. Zhang, T. Zhang, C. Wu, and R. Tao, "Multi-scale spatiotemporal feature fusion network for video saliency prediction," *IEEE Trans. Multimedia*, vol. 26, pp. 4183–4193, 2023.
[47] Y. Zhang, Y. Liu, and C. Wu, "Attention-guided multi-granularity fusion model for video summarization," *Expert Syst. Appl.*, vol. 249, Sep. 2024, Art. no. 123568. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0957417424004330
[48] S. Yi, X. Liu, J. Li, and L. Chen, "UAVformer: A composite transformer network for urban scene segmentation of UAV images," *Pattern Recognit.*, vol. 133, Jan. 2023, Art. no. 109019.
[49] Q. Diao, Y. Dai, C. Zhang, Y. Wu, X. Feng, and F. Pan, "Superpixel-based attention graph neural network for semantic segmentation in aerial images," *Remote Sens.*, vol. 14, no. 2, p. 305, Jan. 2022. [Online]. Available: https://www.mdpi.com/2072-4292/14/2/305
[50] S. Ouyang and Y. Li, "Combining deep semantic segmentation network and graph convolutional neural network for semantic segmentation of remote sensing imagery," *Remote Sens.*, vol. 13, no. 1, p. 119, Dec. 2020. [Online]. Available: https://www.mdpi.com/2072-4292/13/1/119
[51] Y. Zan, P. Lu, and T. Meng, "A gradual adversarial training method for semantic segmentation," *Remote Sens.*, vol. 16, no. 22, p. 4277, Nov. 2024. [Online]. Available: https://www.mdpi.com/2072-4292/16/22/4277
[52] J. Zeng et al., "Unsupervised domain adaptation for remote sensing semantic segmentation with the 2D discrete wavelet transform," *Sci. Rep.*, vol. 14, no. 1, Oct. 2024, Art. no. 23552. [Online]. Available: https://api.semanticscholar.org/CorpusID:273230157
[53] A. Gu et al., "Combining recurrent, convolutional, and continuous-time models with linear state space layers," in *Proc. Adv. neural Inf. Process. Syst.*, vol. 34, 2021, pp. 572–585.
[54] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," 2021, *arXiv:2111.00396*.
[55] A. Gu, A. Gupta, K. Goel, and C. Ré, "On the parameterization and initialization of diagonal state space models," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2022, pp. 35971–35983.
[56] A. Gupta, A. Gu, and J. Berant, "Diagonal state spaces are as effective as structured state spaces," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2022, pp. 22982–22994.
[57] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.
[58] Q. Zhu et al., "Samba: Semantic segmentation of remotely sensed images with state space model," 2024, *arXiv:2404.01705*.
[59] L. Gao, Y. Liu, X. Chen, Y. Liu, S. Yan, and M. Zhang, "CUS3D: A new comprehensive urban-scale semantic-segmentation 3D benchmark dataset," *Remote Sens.*, vol. 16, no. 6, p. 1079, Mar. 2024.
[60] Y. Wang, X. Chen, L. Cao, W. Huang, F. Sun, and Y. Wang, "Multimodal token fusion for vision transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 12186–12195.
[61] J. Zhang, H. Liu, K. Yang, X. Hu, R. Liu, and R. Stiefelhagen, "CMX: Cross-modal fusion for RGB-X semantic segmentation with transformers," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 12, pp. 14679–14694, Dec. 2023.
[62] J. Zhang et al., "Delivering arbitrary-modal semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 1136–1147.
[63] E. Xie et al., "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Sys. (NIPS)*, vol. 34, Dec. 2021, pp. 12077–12090.
[64] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, Jan. 2019, pp. 8024–8035.
[65] Rockchip Semiconductor. *Rk3588 Technical Reference Manual*. Accessed: Apr. 25, 2025. [Online]. Available: https://www.rockchip.com/document-center/product-manuals/3588/



Lin Gao received the B.S. degree in transportation engineering from Southwest Jiaotong University (SWJTU), Chengdu, China, in 2022, and the M.S. degree in management science and engineering from the National University of Defense Technology, Changsha, China, in July 2025.

He is currently employed at AVIC Digital Corporation Ltd., Beijing, China. His research interests include 3-D semantic segmentation.



Xia Li received the B.S. degree in computer science and technology from Taiyuan University of Technology, Taiyuan, China, in 2024. She is currently pursuing the M.S. degree in management science and engineering from the National University of Defense Technology, Changsha, China.

Her research interests include semantic segmentation and visual localization.



Yu Liu received the B.S. degree from Northwestern Polytechnical University, Xi'an, China, in 2005, and the M.Sc. degree in image processing and the Ph.D. degree in computer graphics from the University of East Anglia, Norwich, U.K., in 2007 and 2011, respectively.

He is currently a Professor with the Department of System Engineering, National University of Defense Technology, Changsha, China. His research interests include image/video processing, computer graphics, and visual haptic technology.



Xi Chen received the B.S. degree in electronic information engineering and the M.S. degree in information and communication engineering from Dalian Maritime University, Dalian, China, in 2020 and 2023, respectively. He is currently pursuing the Ph.D. degree with the College of Systems Engineering, National University of Defense Technology, Changsha, China.

His research interests include hyperspectral image processing and deep learning.



Yuxiang Liu received the B.S. degree in information management and information system from Tianjin University of Technology (TJUT), Tianjin, China, in 2016, and the M.S. degree in control science and engineering from the National University of Defense Technology, Changsha, China, in 2020, where he is currently pursuing the Ph.D. degree.

His research interests include thermal imaging, multispectral matching, and 3-D thermal reconstruction.



Jiazhen Xu received the B.S. degree in electronic science and technology from Changsha University of Science and Technology, Changsha, China, in 2020. He is currently pursuing the M.S. degree with the College of Systems Engineering, National University of Defense Technology, Changsha.

His research interests include semantic segmentation, building height estimation, and machine learning.



Maojun Zhang received the B.S. and Ph.D. degrees in system engineering from the National University of Defense Technology, Changsha, China, in 1992 and 1997, respectively.

He is currently a Professor with the Department of System Engineering, National University of Defense Technology. His research interests include computer vision, information system engineering, and system simulation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 08:20:32 UTC from IEEE Xplore. Restrictions apply.